Notes For Hall Algebra

Hanlin Hao

April 24, 2026

Contents

1 Hall Algebras **2**

2 Green's Coproduct **6**

3 Quantum Group-$\mathbf{U}_q(\mathfrak{sl}_2)$ **10**

4 A Geometric Constructioins of $\mathbf{U}_q(\mathfrak{sl}_2)$ **15**

5 Quiver Representations **20**
5.1 Quiver 20
5.2 Representations of Quivers of Finite Type 22

6 Ringel-Green Theorem **27**

7 PBW Theorem **31**
7.1 Braiding Group 31
7.2 PBW Theorem 33

8 Drinfield Double **34**
8.1 Hopf Algebra 34
8.2 Bicrossed Products 36
8.3 Representation of Drinfield Double 38

9 The representation of quivers over $\mathbb{F}_1$ **40**
9.1 The category $\mathrm{Vect}(\mathbb{F}_1)$ 40
9.2 Representation over $\mathbb{F}_1$ 41
9.3 The Hall algebra of $\mathrm{Rep}(Q, \mathbb{F}_1)$ 42

10 Jordan quiver **47**
10.1 Hall algebra of Jordan quiver 47
10.2 Symmetry functions 49
10.3 Irreducible Representations of $\mathbf{U}_q(\mathfrak{sl}_2)$ 51

1 Hall Algebras

In this section, we will introduce some definitions and examples of Hall algebras.

Definition 1.1 (Finitary abelian category)**.** A small category $\mathcal{A}$ is called **finitary** if the following two conditions are satisfied:

i). For any two objects $M, N \in \mathrm{Obj}(\mathcal{A})$, we have $|\mathrm{Hom}(M, N)| < \infty$;

ii). For any two objects $M, N \in \mathrm{Obj}(\mathcal{A})$, we have $|\mathrm{Ext}^1(M, N)| < \infty$.

Definition 1.2 (Grothendieck Group)**.** For an abelian category, we define its Grothendieck group $K(\mathcal{A})$ to be the abelian group generated by symbols $[M]$ with $M \in \mathrm{Obj}(\mathcal{A})$ with the relation

$$[M] - [N] - [L] = 0$$

if $0 \longrightarrow L \longrightarrow M \longrightarrow N \longrightarrow 0$ is a short exact sequence over $\mathcal{A}$.

Proposition 1.3. *If $\mathcal{A}$ is a finite length category, then $K(\mathcal{A})$ is freely generated by the classes of the simple objects.*

Proof. Recall that $\mathcal{A}$ is a finite length category if every object of $\mathcal{A}$ admits a finite composition sequence with simple factors. □

Definition 1.4. The projective dimension of an object $M \in \mathcal{A}$ is the minimal length of a projective resolution a M, denoted by $\mathrm{pd}(M)$. The global dimension of M is defined to be

$$\mathrm{gl.dim}(\mathcal{A}) = \sup_{M \in \mathcal{A}} \mathrm{pd}(M)$$

Example 1.5 (Representations of a quiver Q)**.** For a quiver Q with no oriented cycles, $Rep_k(Q)$ is hereditary, that is, $\mathrm{gl.dim}(Rep_k(Q)) \leq 1$. The simple representations of Q is given by $S(i) = \{V_i = k, V_j = 0\}_{j \in Q}$ and recall that there exits an equivalence between $Rep_k(Q)$ and kQ, the path algebra of Q.

The vertex projective P_i is defined to be $P(i) = \{V_i = ke_i, V_j = k\{\alpha \in Q_1 \big| \alpha : i \to j\}\}_{j \in Q_1}$ and the morphisms between V_s,V_t are given by the product in the loop algebra. Then there exists a projective resolution of $S(i)$:

$$0 \longrightarrow \bigoplus_{\alpha : i \longrightarrow j} P(j) \longrightarrow P(i) \longrightarrow S(i) \longrightarrow 0$$

Hence, for any simple representation S_i, we have $\mathrm{pd}(S_i) \leq 1$ and thus, $\mathrm{gl.dim}(Rep_k(Q)) \leq 1$.

Example 1.6. $\mathrm{gl.dim}(\mathrm{Coh}(\mathbb{P}^1)) = 1$: For $\mathcal{F}, \mathcal{G} \in \mathrm{Coh}(\mathbb{P}^1)$,

$$\mathrm{Ext}^i(\mathcal{F}, \mathcal{G}) = H^i(\mathcal{F} \otimes \mathcal{G}^\vee \otimes \mathcal{O}(-2))^\vee$$

Thus, $\mathrm{Ext}^i(\mathcal{F}, \mathcal{G}) = 0$ when $i > 1$ by the Grothendieck's vanishing theorem.

Let $\mathcal{A}$ be a finitary category and assume that $\mathrm{gl.dim}(\mathcal{A}) < \infty$. For any $M, N \in \mathrm{Ob}(\mathcal{A})$, define

$$\langle M, N \rangle_m = \left(\prod_{i=0}^{\infty} |Ext^i(M, N)|^{(-1)^i} \right)^{\frac{1}{2}}$$

Recall that for any $0 \longrightarrow L \longrightarrow M \longrightarrow K \longrightarrow 0$, applying $\mathrm{Hom}(-, N)$, there exists a long exact sequence

$$\cdots \longrightarrow \mathrm{Ext}^{i-1}(L, N) \longrightarrow \mathrm{Ext}^i(K, N) \longrightarrow \mathrm{Ext}^i(M, N) \longrightarrow \mathrm{Ext}^i(L, N) \longrightarrow \mathrm{Ext}^{i+1}(K, N) \longrightarrow \cdots$$

Then we can see that $\langle M,N\rangle_m = \langle K,N\rangle_m \langle L,N\rangle_m$. Thus, we can define $\langle -,-\rangle_m$ over $K(\mathcal{A})$:

$$\begin{aligned}\langle -,-\rangle_m : K(\mathcal{A}) \times K(\mathcal{A}) &\longrightarrow \mathbb{C} \\ ([M],[N]) &\longmapsto \langle M,N\rangle_m\end{aligned}$$

which is called the **Euler form**.

Let $\chi = \mathrm{Ob}(\mathcal{A})/\sim$ be the set of isomorphism classes of objects in $\mathcal{A}$. Consider

$$\mathbf{H}_{\mathcal{A}} := \bigoplus_{M\in\chi} \mathbb{C}[M]$$

linearly spanned by symbols $[M]$, where M runs through χ. Let $\mathcal{P}^R_{M,N}$ be the set of short exact sequence $0 \longrightarrow N \longrightarrow R \longrightarrow M \longrightarrow 0$ and put $\mathbf{P}^R_{M,N} = |\mathcal{P}^R_{M,N}|$, which is finite since $\mathrm{Hom}(N,R), \mathrm{Hom}(R,M)$ are finite. For each $P \in \mathrm{Obj}(P)$, let $a_P = |\mathrm{Aut}(P)|$.

Proposition 1.7. *The product*

$$[M]\cdot[N] := \langle M,N\rangle_m \sum_R \frac{1}{a_M a_N}\mathbf{P}^R_{M,N}[R]$$

assigns the structure of an associative algebra over $\mathbf{H}_{\mathcal{A}}$. *The unit* $i : \mathbb{C} \longrightarrow \mathbf{H}_{\mathcal{A}}$ *is given by* $i(c) = c[0]$, *where* 0 *is the zero object of* $\mathcal{A}$.

To understand $\frac{1}{a_M a_N}\mathbf{P}^R_{M,N}$, we have the following lemma

Lemma 1.8. *For any* $M,N,R \in \mathrm{Ob}(\mathcal{A})$, *we have*

$$\frac{1}{a_M a_N}\mathbf{P}^R_{M,N} = |\{L \subset R \big| L \cong N \text{ and } R/L \cong M\}|$$

Proof. $\mathrm{Aut}(M)\times\mathrm{Aut}(N)$ acts freely on $\mathcal{P}^R_{M,N}$ and the quotient is $\{L \subset R \big| L \cong N \text{ and } R/L \cong M\}$:

For any $(\varphi,\psi) \in \mathrm{Aut}(M)\times\mathrm{Aut}(N)$,

$$(\varphi,\psi)\cdot[0 \longrightarrow N \xrightarrow{i} R \xrightarrow{p} M \longrightarrow 0] = [0 \longrightarrow N \xrightarrow{i\circ\varphi} R \xrightarrow{\psi^{-1}\circ p} M \longrightarrow 0]$$

This act is free since i is a monomorphism and p is an epimorphism. □

Proof of 1.7. For any $L,M,N \in \mathrm{Ob}(\mathcal{A})$, we have

$$\begin{aligned}([L]\cdot[M])\cdot[N] &= \langle L,M\rangle_m \sum \frac{1}{a_L a_M}\mathbf{P}^R_{M,N}[R]\cdot[N] \\ &= \langle L,M\rangle_m \sum \frac{1}{a_L a_M}\mathbf{P}^R_{M,N}\langle R,N\rangle_m \sum_S \frac{1}{a_R a_N}\mathbf{P}^S_{R,N}[S]\end{aligned}$$

To be continued □

Remark. Since if $M \cong N$, then $[M]=[N]$ in $K(\mathcal{A})$. There is a well-defined surjection $p : \chi(\mathcal{A}) \longrightarrow K(\mathcal{A})$ given by $\overline{M} \mapsto [M]$. Thus, the algebra $\mathbf{H}_{\mathcal{A}}$ is naturally graded by the Grothendieck group $K(\mathcal{A})$ of $\mathcal{A}$: We have

$$\mathbf{H}_{\mathcal{A}} = \bigoplus_{\alpha\in K(\mathcal{A})} \mathbf{H}_{\mathcal{A}}[\alpha], \qquad \mathbf{H}_{\mathcal{A}}[\alpha] = \bigoplus_{p(\overline{M})=\alpha} \mathbb{C}[M]$$

Example 1.9. Let $k = \mathbb{F}_q$. Consider the simplest quiver $Q = \bullet$. $Rep_k(Q) = Vect_k$. There is a single simple object $S \cong k$. Then $\mathrm{Hom}(S,S) = k$ and $\mathrm{Ext}^1(S,S) = \{0\}$. Then

$$\langle S,S\rangle_m = \left(|\mathrm{Hom}(S,S)|^{(-1)^0}\right)^{\frac{1}{2}} = q^{\frac{1}{2}} =: v$$

Then

$$\begin{aligned}
[S^{\oplus i}]\cdot[S^{\oplus j}] &= \left\langle S^{\oplus i}, S^{\oplus j}\right\rangle_m \sum_R \#\{L\subset R \big| L\cong S^{\oplus i} \text{ and } R/L\cong S^{\oplus j}\}[R] \\
&= \left\langle S^{\oplus i}, S^{\oplus j}\right\rangle_m \#\{L\subset S^{\oplus i+j} \big| L\cong S^{\oplus i}\}[S^{\oplus i+j}] \\
&= v^{ij}\#Gr(i,i+j)[S^{\oplus i+j}] \\
&= v^{ij}\begin{bmatrix} i+j \\ i \end{bmatrix}_+ [S^{\oplus i+j}]
\end{aligned}$$

where

$$\begin{bmatrix} i+j \\ i \end{bmatrix}_+ = \prod_{k=0}^{i-1} \frac{q^{i+j-k}-1}{q^{i-k}-1}$$

Now, to see that the product defined in Hall algebra is associated. We have

$$\begin{aligned}
[S^{\oplus k}]\cdot([S^{\oplus i}]\cdot[S^{\oplus j}]) &= v^{ij}\begin{bmatrix} i+j \\ i \end{bmatrix}_+ [S^{\oplus k}]\cdot[S^{\oplus i+j}] \\
&= \langle S^{\oplus k}, S^{\oplus i+j}\rangle \begin{bmatrix} i+j+k \\ k \end{bmatrix}_+ v^{ij}\begin{bmatrix} i+j \\ i \end{bmatrix}_+ [S^{\oplus i+j+k}] \\
&= v^{ij+ik+jk}\begin{bmatrix} i+j \\ i \end{bmatrix}_+ \begin{bmatrix} i+j+k \\ k \end{bmatrix}_+ [S^{\oplus i+j+k}]
\end{aligned}$$

and

$$\begin{aligned}
([S^{\oplus k}]\cdot[S^{\oplus i}])\cdot[S^{\oplus j}] &= v^{ik}\begin{bmatrix} k+i \\ k \end{bmatrix}_+ [S^{\oplus k+i}]\cdot[S^{\oplus j}] \\
&= v^{ij+kj+ik}\begin{bmatrix} k+i \\ k \end{bmatrix}_+ \begin{bmatrix} k+i+j \\ j \end{bmatrix}_+ [S^{\oplus i+j+k}]
\end{aligned}$$

Note that $\mathrm{Gr}(j, k+i+j) \cong \mathrm{Gr}(k+i, k+i+j)$. Since $|Gr(n+m, n+m+k)||Gr(n, m+n)| = |Gr(n+k, n+m+k)||Gr(k, n+k)|$. We can see that the hall algebra defined on Q is associative.

Example 1.10. (If you have already known some basic result in representation of A_2) For A_2 quiver, we know that all its representations can be written as the direct sum of the following three types

$$S_{(1,0)} : \mathbb{F}_p \longrightarrow 0;\ S_{(0,1)} : 0 \longrightarrow \mathbb{F}_p;\ M_{12} : \mathbb{F}_p \xrightarrow{id} \mathbb{F}_p$$

Here $S_{(1,0)}$ and $S_{(0,1)}$ are simple and M_{12} is indecomposable.

We know that $\mathrm{Ext}^1(S_{(1,0)}, S_{(0,1)}) = \mathbb{F}_p$ (because $0 \to S_{(0,1)} \to M_{12} \to S_{(1,0)} \to 0$ is only non-trivial short exact sequence up to isomorphism). So we can compute

$$\begin{aligned}
[S_{(1,0)}^{\oplus i}]\cdot[S_{(0,1)}^{\oplus j}] &= \langle S_{(1,0)}^{\oplus i}, S_{(0,1)}^{\oplus j}\rangle_m \sum_R \#\{L\subset R \big| L\cong S_{(0,1)}^{\oplus i} \text{ and } R/L\cong S_{(1,0)}^{\oplus j}\}[R] \\
&= v^{-ij}\sum_R \#\{L\subset R \big| L\cong S_{(0,1)}^{\oplus i} \text{ and } R/L\cong S_{(1,0)}^{\oplus j}\}[R]
\end{aligned}$$

Then the possible R is given by

$$R = M_{12}^{\oplus k} \oplus S_{(1,0)}^{\oplus i-k} \oplus S_{(0,1)}^{\oplus j-k}$$

with $0 \le k \le \min\{i, j\}$.

Now, let's first compute the case when $i = 1, j = 1$. Then

$$[S_{(1,0)}]\cdot[S_{(0,1)}] = v^{-1}([S_{(1,0)} \oplus S_{(0,1)}] + [M_{12}])$$

since M_{12} has only 1 subspaces that are isomorphic to $S_{(0,1)}$.

Also, note that $\mathrm{Ext}^1(S_{(0,1)}, S_{(1,0)}) = 0$. By very easy computation,

$$[S_{(0,1)}] \cdot [S_{(1,0)}] = [S_{(0,1)} \oplus S_{(1,0)}]$$

which implies $(\mathbf{H}_{\mathcal{A}}, \cdot)$ is **not** necessarily a commutative algebra.

2 Green's Coproduct

In this section, we want to understand how to upgrade the hall algebra to a bialgebra.

Recall that the Hall algebra $\mathbf{H}_{\mathcal{A}}$ of $\mathcal{A}$ is graded by the Grothendieck group. Hence, we can define

$$\mathbf{H}_{\mathcal{A}}[\alpha]\hat{\otimes}\mathbf{H}_{\mathcal{A}}[\beta] = \prod_{[M]=\alpha,[N]=\beta} \mathbb{C}[M]\otimes\mathbb{C}[N]$$

$$\mathbf{H}_{\mathcal{A}}\hat{\otimes}\mathbf{H}_{\mathcal{A}} = \prod_{\alpha,\beta\in K(\mathcal{A})} \mathbf{H}_{\mathcal{A}}[\alpha]\hat{\otimes}\mathbf{H}_{\mathcal{A}}[\beta]$$

Then, we define the coproduct and the counit

$$\Delta([R]) = \sum_{M,N} \langle M,N\rangle_m \frac{1}{a_R}\mathbf{P}^R_{M,N}[M]\otimes[N]$$

$$\epsilon : \mathbf{H}_{\mathcal{A}} \longrightarrow \mathbb{C} \qquad [M]\longmapsto \delta_{M,0}$$

Proposition 2.1. *Δ and ϵ define a structure of topological associated coproduct on $\mathbf{H}_{\mathcal{A}}$.*

Proof. We need to check that $(\Delta\otimes\mathrm{Id})\circ\Delta = (\mathrm{Id}\otimes\Delta)\circ\Delta$.

$$\begin{aligned}(\Delta\otimes\mathrm{Id})\circ\Delta([R]) &= \sum_{M,N}\langle M,N\rangle_m \frac{1}{|a_R|}\mathbf{P}^R_{M,N}\Delta([M])\otimes N\\ &= \sum_{M,N}\frac{1}{|a_R|}\mathbf{P}^R_{M,N}\sum_{S,Q}\langle S,Q\rangle_m [S]\otimes[Q]\otimes[N]\end{aligned}$$

$$\begin{aligned}(\mathrm{Id}\otimes\Delta)\circ\Delta([R]) &= \sum_{M,N}\langle M,N\rangle_m \frac{1}{|a_R|}\mathbf{P}^R_{M,N}[M]\otimes\Delta([N])\\ &= \sum_{M,N}\langle M,N\rangle_m \frac{1}{|a_R|}\mathbf{P}^R_{M,N}\sum_{S,Q}\langle S,Q\rangle_m [M]\otimes[S]\otimes[Q]\\ &= \sum_{S,T}\langle S,T\rangle_m \frac{1}{|a_R|}\mathbf{P}^R_{S,T}\sum_{Q,N}\langle Q,N\rangle_m \mathbf{P}^T_{Q,N}[S]\otimes[Q]\otimes[N]\end{aligned}$$

where the last equality is gained by replacing (M,S,Q,N) by (S,Q,N,T). Finally, we just check

$$\sum_T \langle S,T\rangle_m\cdot\langle Q,N\rangle_m \frac{\mathbf{P}^R_{S,T}\mathbf{P}^T_{Q,N}}{|a_R||a_T|} = \sum_M \langle M,N\rangle_m\langle S,Q\rangle_m \frac{\mathbf{P}^P_{M,N}\mathbf{P}^M_{S,Q}}{|a_R||a_M|}$$

We can show that when $\mathcal{A}$ is hereditary, and if $0\longrightarrow S\longrightarrow M\longrightarrow Q\longrightarrow 0$, then using the long exact sequence $\langle M,C\rangle_m = \langle S,C\rangle_m\langle Q,C\rangle_m$. Thus, writing $\langle S,T\rangle_m = \langle S,Q\rangle_m\langle S,N\rangle_m$ and $\langle M,N\rangle_m = \langle S,N\rangle_m\langle Q,N\rangle_m$, the equality above becomes

$$\sum_T \frac{1}{a_T}\mathbf{P}^R_{S,T}\mathbf{P}^T_{Q,N} = \sum_M \frac{1}{a_M}\mathbf{P}^R_{M,N}\mathbf{R}^M_{S,Q}$$

We can use stack to prove this formula. **To be continued.** □

Remark 2.2. Δ doesn't defines a coproduct on $\mathbf{H}_{\mathcal{A}}$, it is in fact a morphism $\mathbf{H}_{\mathcal{A}}\longrightarrow\mathbf{H}_{\mathcal{A}}\hat{\otimes}\mathbf{H}_{\mathcal{A}}$. That's why we call it topologically.

Example 2.3. (a). In 1.9, $\langle S,S\rangle = q^{\frac{1}{2}}$ and let $v = \langle S,S\rangle_m$. Then

$$\begin{aligned}\Delta([S^{\oplus n}]) &= \sum_{M,N} \langle M,N\rangle_m \frac{1}{|a_{S^{\oplus n}}|}\mathbf{P}^{S^{\oplus n}}_{M,N}[M]\otimes[N] \\ &= \sum_{0\le k\le n} \left\langle S^{\oplus k}, S^{\oplus n-k}\right\rangle_m \frac{1}{\#\mathrm{GL}(n;\mathbb{F}_q)}\#\mathrm{GL}(k;\mathbb{F}_q)\#\mathrm{GL}(n-k;\mathbb{F}_q)\#\mathrm{Gr}(n,k)[S^{\oplus k}]\otimes[S^{\oplus n-k}] \\ &= \sum_{0\le k\le n} v^{-(n-k)k}[S^{\oplus k}]\otimes[S^{\oplus n-k}]\end{aligned}$$

Note that

$$\#\mathrm{GL}(k;\mathbb{F}_q) = \prod_{0\le t\le k-1}(p^k - p^t)$$

Example 2.4. For the representations of A_2 over F_p.

$$\begin{aligned}\Delta([M_{12}]) &= \sum_{M,N}\langle M,N\rangle_m \frac{1}{a_{M_{12}}}\mathbf{P}^{M_{12}}_{M,N}[M]\otimes[N] \\ &= 1\otimes[M] + [M]\otimes 1 + v^{-1}(p-1)[S_{(1,0)}]\otimes[S_{(0,1)}]\end{aligned}$$

Here, $0\to S_{(0,1)}\to M_{12}\to S_{(1,0)}\to 0$. So by

$$\frac{1}{a_{S_{(1,0)}}a_{S_{(0,1)}}}\mathbf{P}^{M_{12}}_{S_{(1,0)},S_{(0,1)}} = \#\{R\subset M_{12}\big|R\cong S_{(0,1)}, M_{12}/R\cong S_{(1,0)}\} = 1$$

Thus, $\mathbf{P}^{M_{12}}_{S_{(1,0)},S_{(0,1)}} = (p-1)^2$.

Remark 2.5. Here are some properties of $\Delta : \mathbf{H}_{\mathcal{A}} \longrightarrow \mathbf{H}_{\mathcal{A}}\hat{\otimes}\mathbf{H}_{\mathcal{A}}$:

(i). The map Δ respects the grading by $K(\mathcal{A})$, that is

$$\Delta(\mathbf{H}_{\mathcal{A}}[\gamma]) \subset \prod_{\alpha+\beta=\gamma}\mathbf{H}_{\mathcal{A}}[\alpha]\hat{\otimes}\mathbf{H}_{\mathcal{A}}[\beta]$$

ii). For S a simple object, we have $\Delta([S]) = [S]\otimes 1 + 1\otimes[S]$, that is, $[S]$ is a primitive element of $\mathbf{H}_{\mathcal{A}}$.

iii). The coproduct may not be cocommutative. Let's just consider 2.4, since $[S_{(1,0)}]\otimes[S_{(0,1)}]\neq[S_{(0,1)}]\otimes[S_{(1,0)}]$.

iv). As Δ is coassociative, we may consider an iterated comultiplication $\Delta^r : \mathbf{H}_{\mathcal{A}}\to\mathbf{H}_{\mathcal{A}}\hat{\otimes}\cdots\hat{\otimes}\mathbf{H}_{\mathcal{A}}$, which has the following interpretation:

$$\Delta^r([R]) = \sum_{M_1,\dots,M_r}\Big(\prod_{i<j}\langle M_i,M_j\rangle\frac{a_{M_1}\cdots a_{M_r}}{a_R}\times\#\{L_r\subset\cdots\subset L_1 = R\big|L_i/L_{i+1}\cong M_i\}[M_1]\otimes\cdots\otimes[M_r]$$

v). The coproduct Δ takes values in $\mathbf{H}_{\mathcal{A}}\otimes\mathbf{H}_{\mathcal{A}}$ if and only if the following condition holds: Any fixed object R of $\mathcal{A}$ has only finitely many subobjects $N\subset R$, called **finite subobjects condition**.

For $\gamma\in K(\mathcal{A})$, let

$$\mathbf{1}_\gamma = \sum_{M=\gamma}[M]$$

be the sum of all objects in $\mathcal{A}$ of class γ.

Lemma 2.6. *Assume that* $\mathrm{gldim}(\mathcal{A})\le 1$. *Then we have*

$$\Delta(\mathbf{1}_\gamma) = \sum_{\alpha+\beta=\gamma}\langle\alpha,\beta\rangle_m^{-1}\mathbf{1}_\alpha\otimes\mathbf{1}_\beta$$

Example 2.7. To be continued

If x, y, z, w are homogeneous elements in $\mathbf{H}_{\mathcal{A}}$ of respective weight $\mathrm{wt}(x), \mathrm{wt}(y), \mathrm{wt}(z), \mathrm{wt}(w)$ in $K(\mathcal{A})$, then we define a new multiplication over $\mathbf{H}_{\mathcal{A}} \otimes \mathbf{H}_{\mathcal{A}}$ by

$$(x \otimes y) * (z \otimes w) = (\mathrm{wt}(y), \mathrm{wt}(z))_m (xz \otimes yw)$$

called **the twisted multiplication**.

Now, we have a morphism $\Delta : \mathbf{H}_{\mathcal{A}} \to \mathbf{H}_{\mathcal{A}} \otimes \mathbf{H}_{\mathcal{A}}$. We want to find the what will happen if we combine Δ with $*$? To begin with, we see that the multiplication map over $\mathbf{H}_{\mathcal{A}} \otimes \mathbf{H}_{\mathcal{A}}$ doesn't extend to the completion. Then, we say a product

$$(\sum_i a_i \otimes b_i) * (\sum_j c_j \otimes d_j)$$

of elements of $\mathbf{H}_{\mathcal{A}} \hat{\otimes} \mathbf{H}_{\mathcal{A}}$ is convergent if for any pair of objects R, S in $\mathcal{A}$, the coefficient of $[R] \otimes [S]$ in $(a_i \otimes b_i) * (c_j \otimes d_j)$ is nonzero for only finitely many values of (i, j). **In this case**, the product is well-defined in $\mathbf{H}_{\mathcal{A}} \hat{\otimes} \mathbf{H}_{\mathcal{A}}$.

Lemma 2.8. *Let M, N be objects in $\mathcal{A}$. Then the product $\Delta([M]) * \Delta([N])$ converges.*

Theorem 2.9 (Green)**.** *The map $\Delta : \mathbf{H}_{\mathcal{A}} \to \mathbf{H}_{\mathcal{A}} \hat{\otimes} \mathbf{H}_{\mathcal{A}}$ is a morphism of algebras, that is, for any $x, y \in \mathbf{H}_{\mathcal{A}}$, we have $\Delta(x \cdot y) = \Delta(x) * \Delta(y)$.*

Example 2.10. Now, we just discuss Δ for $\mathcal{A} = \mathrm{Vect}_{\mathbb{F}_p}$. Let $x = [S^{\oplus i}]$ and $y = [S^{\oplus j}]$. Then

$$x \cdot y = v^{ij} \begin{bmatrix} i+j \\ i \end{bmatrix}_+ [S^{\oplus i+j}]$$

and

$$\Delta(x \cdot y) = v^{ij} \begin{bmatrix} i+j \\ i \end{bmatrix}_+ \sum_{0 \le k \le i+j} v^{-(i+j-k)k} [S^{\oplus i+j-k}] \otimes [S^{\oplus k}]$$

Now, we compute $\Delta(x)$ and $\Delta(y)$:

$$\Delta(x) = \sum_{0 \le k \le i} v^{-(i-k)k} [S^{\oplus i-k}] \otimes [S^{\otimes k}]; \Delta(y) = \sum_{0 \le k \le j} v^{-(j-k)k} [S^{\oplus j-k}] \otimes [S^{\otimes k}]$$

For any $[S^{\oplus i-k}] \otimes [S^{\oplus k}]$ and $[S^{\oplus j-s}] \otimes [S^{\otimes s}]$, we can compute

$$\begin{aligned} ([S^{\oplus i-k}] \otimes [S^{\oplus k}]) * ([S^{\oplus j-s}] \otimes [S^{\oplus s}]) &= v^{k(j-s)+(i-k)(j-s)+ks} \begin{bmatrix} i+j-s-k \\ i-k \end{bmatrix}_+ \begin{bmatrix} k+s \\ k \end{bmatrix}_+ [S^{\oplus i+j-s-k}] \otimes [S^{\oplus s+k}] \\ &= v^{ij+ks-is} \dots \end{aligned}$$

Now, let $n = s + k$, then we can see that

$$\Delta(x) * \Delta(y) = \sum_{0 \le n \le i+j} \sum_k v^{ij+k(n-k)-i(n-k)} \begin{bmatrix} i+j-n \\ i-k \end{bmatrix}_+ \begin{bmatrix} n \\ k \end{bmatrix}_+ [S^{\oplus i+j-n}] \otimes [S^{\oplus n}]$$

To prove that $\Delta(x) * \Delta(y) = \Delta(x \cdot y)$, we need **q-Chu-Vandermonde identity**:

$$\begin{bmatrix} m+n \\ r \end{bmatrix}_+ = \sum_k v^{k(m-r)} \begin{bmatrix} n \\ r-k \end{bmatrix}_+ \begin{bmatrix} m \\ r \end{bmatrix}_+$$

Let $\mathbf{K} = \mathbb{C}[K(\mathcal{A})]$ and $\mathbf{k}_\alpha$ be the element corresponding to $\alpha \in K(\mathcal{A})$. Then we define **extended Hall-algebra** , $\tilde{\mathbf{H}}_{\mathcal{A}} := \mathbf{H}_{\mathcal{A}} \otimes_{\mathbb{C}} \mathbf{K}$ with the relations

$$\mathbf{k}_\alpha [M] \mathbf{k}_\alpha^{-1} = (\alpha, M)_m [M]$$

Note that $\tilde{\mathbf{H}}_{\mathcal{A}}$ is still $K(\mathcal{A})$-graded where $\deg \mathbf{k}_\alpha = 0$.

Next, we define the comultiplication and counit over $\tilde{\mathbf{H}}_{\mathcal{A}}$. Set the comultiplication

$$\begin{aligned} \Delta : \tilde{\mathbf{H}}_{\mathcal{A}} &\longrightarrow \tilde{\mathbf{H}}_{\mathcal{A}} \hat{\otimes} \tilde{\mathbf{H}}_{\mathcal{A}} \\ \mathbf{k}_\alpha &\longmapsto \mathbf{k}_\alpha \otimes \mathbf{k}_\alpha \\ [R]\mathbf{k}_\alpha &\longmapsto \sum_{M,N} \langle M, N \rangle_m \frac{1}{a_R} \mathbf{P}^R_{M,N} [M] \mathbf{k}_{\bar{N}+\alpha} \otimes [N] \mathbf{k}_\alpha \end{aligned}$$

and the counit

$$\begin{aligned} \varepsilon : \tilde{\mathbf{H}}_{\mathcal{A}} &\longrightarrow \mathbb{C} \\ [M]\mathbf{k}_\alpha &\longmapsto \begin{cases} 0, & \text{if } M \neq 0; \\ 1, & \text{if } M = 0. \end{cases} \end{aligned}$$

Moreover, if defining the multiplication over $\tilde{\mathbf{H}}_{\mathcal{A}} \hat{\otimes} \tilde{\mathbf{H}}_{\mathcal{A}}$ by $(x \otimes y) \cdot (z \otimes w) := xz \otimes yw$, Green's theorem 2.9 can be extended to

Corollary 2.11. *The map* $\Delta : \tilde{\mathbf{H}}_{\mathcal{A}} \longrightarrow \tilde{\mathbf{H}}_{\mathcal{A}} \hat{\otimes} \tilde{\mathbf{H}}_{\mathcal{A}}$

3 Quantum Group-$\mathbf{U}_q(\mathfrak{sl}_2)$

In this section, we will focus on the quantum group $U_v(\mathfrak{sl}_2)$ and study some properties and structures of it.

First of all, let's recall generators and relations over $\mathfrak{sl}_2$. Set

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

with the commutation relations

$$[E, F] = H, \quad [H, E] = 2E, \quad [H, F] = 2F$$

The universal enveloping algebra $\mathbf{U}(\mathfrak{sl}_2)$ of $\mathfrak{sl}_2$ is the associative algebra with generators E, F, H and relations

$$HE - EH = 2E, \quad HF - FH = -2F, \quad EF - FE = H$$

Recall that every representation of sl_2 is determined by its highest weight.

Now, we can replace the generator H by a collection of orthogonal idempotents.

Definition 3.1. The **modified universal enveloping algebra** $\dot{\mathbf{U}}(\mathfrak{sl}_2)$ be the k-algebra defined by

$$\dot{\mathbf{U}}(\mathfrak{sl}_2) = \bigoplus_{\lambda \in \mathbf{Z}} \mathbf{U}(\mathfrak{sl}_2)\mathbf{1}_\lambda$$

with generators:

$$E\mathbf{1}_\lambda, \quad F\mathbf{1}_\lambda, \quad \mathbf{1}_\lambda$$

where $\lambda \in \mathbf{Z}$ and with the relations:

$$\mathbf{1}_\lambda \cdot \mathbf{1}_\mu = \delta_{\lambda\mu}\mathbf{1}_\lambda$$

$$E\mathbf{1}_\lambda = \mathbf{1}_{\lambda+2}E\mathbf{1}_\lambda, \quad F\mathbf{1}_\lambda = \mathbf{1}_{\lambda-2}F\mathbf{1}_\lambda, \quad [E, F]\mathbf{1}_\lambda = \lambda\mathbf{1}_\lambda$$

Example 3.2. Here, we just give some examples to understand the product over $\dot{\mathbf{U}}(\mathfrak{sl}_2)$:

$$(E\mathbf{1}_\lambda) \cdot (F\mathbf{1}_{\lambda+2}) = (E\mathbf{1}_\lambda) \cdot (\mathbf{1}_\lambda F\mathbf{1}_{\lambda+2}) = EF\mathbf{1}_{\lambda+2}$$

$$(F\mathbf{1}_{\lambda+2}) \cdot (E\mathbf{1}_\lambda) = FE\mathbf{1}_\lambda$$

$$(E\mathbf{1}_\lambda + F\mathbf{1}_{\lambda+2}) \cdot (E\mathbf{1}_\lambda - F\mathbf{1}_{\lambda+2}) = FE\mathbf{1}_\lambda - EF\mathbf{1}_{\lambda+2}$$

Definition 3.3. Fix q. We define **quantum integer**:

$$[n]_q = \frac{q^n - q^{-n}}{q - q^{-1}}$$

with $n \in \mathbb{Z}$ and **quantum factorial**:

$$[n]_q! = \prod_{k=1}^{n} [k]_q$$

Moreover, we also define

$$\binom{n}{k}_q = \frac{[n]_q!}{[k]_q![n-k]_q!}$$

When $n \geq 0$, we know that

$$[n]_q = q^{-n+1} + q^{-n+3} + \cdots + q^{n-3} + q^{n-1}$$

For example, $[1]_q = 1$, $[2]_q = q^{-1} + q$, $[3]_q = q^{-2} + 1 + q^2$.

With quantum integers, we can define the modified quantum group of $\mathfrak{sl}_2$:

Definition 3.4. $\dot{\mathbf{U}}_q(\mathfrak{sl}_2)$ is generated by $\{E\mathbf{1}_\lambda, F\mathbf{1}_\lambda, \mathbf{1}_\lambda \big| \lambda \in \mathbb{Z}\}$ with relations:

$$\mathbf{1}_\lambda \cdot \mathbf{1}_\mu = \delta_{\lambda\mu}\mathbf{1}_\lambda$$

$$E\mathbf{1}_\lambda = \mathbf{1}_{\lambda+2}E\mathbf{1}_\lambda, \quad F\mathbf{1}_\lambda = \mathbf{1}_{\lambda-2}F\mathbf{1}_\lambda, \quad [E,F]\mathbf{1}_\lambda = [\lambda]_q\mathbf{1}_\lambda$$

Now, we define the quantum group, which is the deformation of some enveloping algebra. To begin with, we first define the quantum group of $\mathfrak{sl}_2$:

Definition 3.5. The **quantum group** $\mathbf{U}_q(\mathfrak{sl}_2)$ is an algebra over $\mathbb{Q}(q)$ of rational functions given by generators E, F, K, K^{-1} and relations

$$K \cdot K^{-1} = K^{-1} \cdot K = 1 \tag{R1}$$

$$KE = q^2 EK \tag{R2}$$

$$KF = q^{-2} FK \tag{R3}$$

$$EF - FE = \frac{K - K^{-1}}{q - q^{-1}} \tag{R4}$$

Remark. We can assign $\mathbf{U}_q(\mathfrak{sl}_2)$ with a graded structure:

$$\deg E = 1, \quad \deg F = -1, \quad \deg K = \deg K^{-1} = 0$$

Now, we'll show that $\mathbf{U}_q(\mathfrak{sl}_2)$ admits a structure of Hopf algebra.

Lemma 3.6. *There exists unique homomorphism of $\mathbb{Q}(q)$-algebras $\Delta : U \longrightarrow U \otimes U$ given by*

$$\Delta(E) = E \otimes 1 + K \otimes E \tag{3.1}$$

$$\Delta(F) = F \otimes K^{-1} + 1 \otimes F \tag{3.2}$$

$$\Delta(K) = K \otimes K \tag{3.3}$$

Proof. Need to show that $(E \otimes 1 + K \otimes E, F \otimes K^{-1} + 1 \otimes F, K \otimes K, K^{-1} \otimes K^{-1})$ satisfies $(R1) \sim R(4)$. $(R1)$ is immediately from the definition.

Verify $R2$:

$$\begin{aligned}(K \otimes K) \cdot (E \otimes 1 + K \otimes E) &= KE \otimes K + K^2 \otimes KE \\ &= q^2(EK \otimes K + K^2 \otimes EK) \\ &= (E \otimes 1 + K \otimes E) \cdot (K \otimes K)\end{aligned}$$

and $R3$ is verified in the same way immediately.

Now, for $R4$,

$$(E \otimes 1 + K \otimes E) \cdot (F \otimes K^{-1} + 1 \otimes F) = EF \otimes K^{-1} + E \otimes F + KF \otimes EK^{-1} + K \otimes EF$$
$$(F \otimes K^{-1} + 1 \otimes F) \cdot (E \otimes 1 + K \otimes E) = FE \otimes K^{-1} + FK \otimes K^{-1}E + E \otimes F + K \otimes FE$$

Thus,

$$[E \otimes 1 + K \otimes E, F \otimes K^{-1} + 1 \otimes F] = [E,F] \otimes K^{-1} + K \otimes [E,F] = \frac{K \otimes K - K^{-1} \otimes K^{-1}}{q - q^{-1}}$$

Note, here we use $KF \otimes EK^{-1} = q^{-2}FK \otimes q^2K^{-1}E = FK \otimes K^{-1}E$. $\square$

Proposition 3.7.

$$\Delta(K^n) = K^n \otimes K^n$$
$$\Delta(E^r) = \sum_{i=1}^{r} q^{i(r-i)} \binom{r}{i}_q E^{r-i}K^i \otimes E^i$$
$$\Delta(F^r) = \sum_{i=0}^{r} q^{i(r-i)} \binom{r}{i}_q F^i \otimes F^{r-i}K^{-i}$$

Proof. The first identity is obvious and the last two are proved by induction on r, since $r = 1$ holds by 3.6.

$$\begin{aligned}
&(\sum_{i=0}^{r-1} q^{i(r-1-i)} \binom{r-1}{i}_q E^{r-1-i}K^i \otimes E^i) \cdot (E \otimes 1 + K \otimes E) \\
=&\sum_{i=0}^{r-1} q^{i(r-i-i)} \binom{r-1}{i}_q E^{r-1-i}K^iE \otimes E^i + \sum_{i=0}^{r-1} q^{i(r-1-i)} \binom{r-1}{i}_q E^{r-1-i}K^{i+1} \otimes E^{i+1} \\
=&E^{r-1} \otimes 1 + K^r \otimes E^r + \sum_{i=1}^{r-1} (q^{i(r-1-i)} \binom{r-1}{i}_q E^{r-1-i}K^iE \otimes E^i + q^{(i-1)(r-i)} \binom{r-1}{i-1}_q E^{r-i}K^i \otimes E^i \\
=&E^{r-1} \otimes 1 + K^r \otimes E^r + \sum_{i=1}^{r-1} (q^{i(r-i)+i} \binom{r-1}{i}_q + q^{(i-1)(r-i)} \binom{r-1}{i-1}_q) E^{r-i}K^i \otimes E^i
\end{aligned}$$

Then we use the formula

$$q^i \binom{r-1}{i}_q + q^{-(r-i)} \binom{r-1}{i-1}_q = \binom{r}{i}_q$$

See [[Jan96] Chapter 0]. □

We say a morphism is coassociative if the following diagram commutes:

$$\begin{tikzcd}
 & A \otimes A \arrow[r, "1\otimes\Delta"] & A \otimes (A \otimes A) \arrow[dd, "c"] \\
A \arrow[ur, "\Delta"] \arrow[dr, "\Delta"'] & & \\
 & A \otimes A \arrow[r, "\Delta\otimes 1"] & (A \otimes A) \otimes A
\end{tikzcd}$$

where c is the canonical morphism from $A \otimes (A \otimes A) \longrightarrow (A \otimes A) \otimes A$.

Lemma 3.8. *Comultiplication defined on $\mathbf{U}_q(\mathfrak{sl}_2)$ is coassociative.*

Lemma 3.9. *There is a unique homomorphism of $\mathbb{Q}(q)$-algebras $\epsilon : \mathbf{U}_q(\mathfrak{sl}_2) \longrightarrow \mathbb{Q}(q)$ with $\epsilon(E) = \epsilon(F) = 0$ and $\epsilon(K) = 1$ and*

$$\begin{tikzcd}
U \arrow[r, "\Delta"] \arrow[d, "\mathrm{Id}"] & U \otimes U \arrow[d, "1\otimes\epsilon"] \\
U \arrow[r, "c"] & U \otimes \mathbb{Q}(q)
\end{tikzcd}
\qquad
\begin{tikzcd}
U \arrow[r, "\Delta"] \arrow[d, "\mathrm{Id}"] & U \otimes U \arrow[d, "\epsilon\otimes 1"] \\
U \arrow[r, "c"] & \mathbb{Q}(q) \otimes U
\end{tikzcd}$$

where $c : U \to U \otimes \mathbb{Q}(q)$ is given by $u \longmapsto u \otimes 1$ and $c : U \to \mathbb{Q}(q) \otimes U$ is given by $u \mapsto 1 \otimes u$.

Proof. Verify that $(0,0,1,1)$ satisfies $(R1) \sim (R4)$ under the multiplication over $\mathbb{Q}(q)$, which is immediately by the definition.

The commutativity of two diagrams is obvious. $\square$

Lemma 3.10. *There os a unique anti-automorphism S of $\mathbf{U}_q(\mathfrak{sl}_2)$ with*

$$S(E) = -K^{-1}E, \quad S(F) = -FK, \quad S(K) = K^{-1}$$

Moreover, one has $S^2(u) = K^{-1}uK$ for all $u \in \mathbf{U}_q(\mathfrak{sl}_2)$.

Proof. We have to show that $(-K^{-1}E, -FK, K^{-1}, K)$ satisfies $(R1) \sim (R4)$ in U^{opp} where we denote the multiplication by $*$.

$$(K^{-1}) * (-K^{-1}E) = (-K^{-1}E)(K^{-1}) = q^2(K^{-1})(-K^{-1}E) = q^2(-K^{-1}E) * (K^{-1})$$

Then similarly, we can verify the result. $\square$

Remark. One has $S(K^r) = K^{-r}$. By induction on r, we can prove:

$$S(E^r) = (-1)^r q^{r(r-1)} K^{-r} E^r, \quad S(F^r) = (-1)^r q^{-r(r-1)} F^r K^r$$

Lemma 3.11. *The following diagrams are commutative:*

$$\begin{tikzcd} U \arrow[r, "\Delta"] \arrow[d, "l\circ\epsilon"] & U \otimes U \arrow[d, "1\otimes S"] \\ U & U \otimes U \arrow[l, "m"] \end{tikzcd} \qquad \begin{tikzcd} U \arrow[r, "\Delta"] \arrow[d, "l\circ\epsilon"] & U \otimes U \arrow[d, "S\otimes 1"] \\ U & U \otimes U \arrow[l, "m"] \end{tikzcd}$$

where $m : U \otimes U \to U$ is given by $u \otimes v \mapsto uv$ and $l : \mathbb{Q}(q) \to U$ is given by $a \mapsto a \cdot 1_U$.

Remark. m, S are not ring homomorphisms! Since $(u \otimes v) \cdot (u' \otimes v') = uu' \otimes vv'$. However, $uu'vv' \neq uvu'v'$.

Proof. We just verify the first diagram for E, F, K, K^{-1}. To begin with, we have

$$\begin{aligned} &E \mapsto E \otimes 1 + K \otimes E \mapsto E \otimes 1 + K \otimes (-K^{-1}E) \mapsto E + K(-K^{-1}E) = 0 \\ &F \mapsto F \otimes K^{-1} + 1 \otimes F \mapsto F \otimes K + 1 \otimes (-FK) \mapsto FK + (-FK) = 0 \\ &K \mapsto K \otimes K \mapsto K \otimes K^{-1} \mapsto KK^{-1} = 1 \end{aligned}$$

However, it is not enough to verify these since m, S are **not** ring homomorphism. Let $f := m \circ (1 \otimes S) \circ \Delta$. Then we what's to verify if $f(uv) = l \circ \epsilon(uv)$ if $f(u) = l \circ \epsilon(u)$ and $f(v) = l \circ \epsilon(v)$: Suppose that $\Delta(u) = \sum u_i \otimes u_i'$ and $\Delta(v) = \sum v_j \otimes v_j'$. Then

$$uv \mapsto \sum u_i v_j \otimes u_i' v_j' \mapsto \sum u_i v_j \otimes S(v_j')S(u_i') \mapsto \sum u_i v_j S(v_j')S(u_i') = \sum u_i f(v) S(u_i) = f(u)f(v)$$

which just equals to $l \circ \epsilon(uv)$ since $l \circ \epsilon(uv)$ is a ring homomorphism. $\square$

To summit, through 3.6, 3.10, 3.9 and 3.11, we assign $\mathbf{U}_q(\mathfrak{sl}_2)$ a structure of Hopf algebra via:

$$\begin{aligned} \Delta(E) &= E \otimes 1 + K \otimes E; \quad \epsilon(E) = 0; \quad S(E) = -K^{-1}E \\ \Delta(F) &= F \otimes K^{-1} + 1 \otimes F; \quad \epsilon(F) = 0; \quad S(F) = -FK \\ \Delta(K) &= K \otimes K; \quad \epsilon(K) = 1; \quad S(K) = K^{-1} \end{aligned}$$

Under the setting above, Δ is not cocommutative under. Since under the swap map $\tau : \mathbf{U}_q(\mathfrak{sl}_2) \otimes \mathbf{U}_q(\mathfrak{sl}_2) \longrightarrow \mathbf{U}_q(\mathfrak{sl}_2) \otimes \mathbf{U}_q(\mathfrak{sl}_2)$,

$$\tau \circ \Delta \neq \Delta$$

However, the comultiplication defined on $\mathbf{U}(\mathfrak{sl}_2)$ is cocommutative:

$$\Delta(x) = x \otimes 1 + 1 \otimes x$$

which satisfies $\tau \circ \Delta = \Delta$.

Now, let's consider the representation of $\mathbf{U}_q(\mathfrak{sl}_2)$.

Since $\mathbf{U}_q(\mathfrak{sl}_2)$ is a Hopf algebra, for any two representation $\rho_i : \mathbf{U}_q(\mathfrak{sl}_2) \otimes V_i \to V_i$, $i = 1, 2$. We can define the tensor product of them via

$$u \cdot (v_1 \otimes v_2) = \Delta(u)(v_1 \otimes v_2) = \sum u_i v_1 \otimes u_i' v_2$$

Example 3.12. Consider $\dim V = 2$ with $V = \mathrm{Span}\{v_+, v_-\}$ and

$$\begin{aligned} Ev_+ = 0; &\quad Ev_- = v+ \\ Ev_+ = v_-; &\quad Ev_-0 \\ Kv_+ = qv_+; &\quad Ev_- = q^{-1}v_- \end{aligned}$$

We can see that under the basis v_+, v_-

$$E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}; \quad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}; \quad H = \begin{pmatrix} q & 0 \\ 0 & q^{-1} \end{pmatrix}$$

we can verify that $E, F, K,^{-1}$ defined above satisfies $(R1) \sim (R4)$. So we get a representation of $\mathbf{U}_q(\mathfrak{sl}_2)$.

Define the swap map on $V \otimes V$:

$$\begin{aligned} \tau : V \otimes V &\longrightarrow V \otimes V \\ u \otimes v &\longmapsto v \otimes u \end{aligned}$$

Observe that

$$\begin{aligned} \tau(E \cdot (v_- \otimes v_+)) = \tau((E \otimes 1 + K \otimes E)(v_- \otimes v_+)) &= v_+ \otimes v_+ \\ E \cdot \tau(v_- \otimes v_+) = (E \otimes 1 + K \otimes E)(v_+ \otimes v_-) &= qv_+ \otimes v_+ \end{aligned}$$

Then we can see that $\tau \circ E \neq E \circ \tau$.

4 A Geometric Constructioins of $\mathrm{U}_q(\mathfrak{sl}_2)$

Fix vector space V_1, V_2 with $\dim V_i = d_i$, $i = 1, 2$. Consider

$$M(d_2, d_1) := \mathrm{Hom}(V_1, V_2) \oplus \mathrm{Hom}(V_2, V_1).$$

Then we get a $\mathrm{GL}(V_2)$ action on $M(d_1, d_2)$.

$$g \cdot (f_1, f_2) := (g \circ f_1, f_2 \circ g^{-1})$$

Define

$$\mu : M(d_1, d_2) \longrightarrow \mathrm{End}(V_2);\ (f_1, f_2) \longmapsto f_1 \circ f_2$$

We say that a point (f_1, f_2) is stable if f_2 is injective. Then we can define the **quiver variety**:

$$\{(f_1, f_2) \in \mu^{-1}(0) \big| (f_1, f_2) \text{ is stable}\}/\mathrm{GL}(V_2)$$

Remark. The stability here corresponding to the stability of representations of quivers: Let $\vec{Q} = (I, H)$ be a quiver (we will define later). For any rep $(V_i)_{i\in I}$, let $\vec{d} = (\dim V_i)_{i\in I}$. Then choose a stability parameter θ such that

$$\sum_{i\in I} \theta_i d_i = 0$$

For any sub-representation $V' \subsetneq V$, denote $d' = (\dim V_i')_{i\in I}$. We say V is **semi-stable** if $\sum_{i\in I} d_i'\theta_i \leq 0$ for all non-trivial V'; V is **stable** if the inequality is strict for all non-trivial V'.

Now, come back to our example. Consider $d = (d_1, d_2)$ and $f = (f_1, f_2)$.

$$V_1 \underset{f_2}{\overset{f_1}{\rightleftarrows}} V_2$$

Then take $\theta = (d_2, -d_1)$. We can see that $(0, \ker f_2)$ is a subrepresentation of (V_1, V_2). Then since we need the representation to be stable, we have

$$-d_1 \dim \ker f_2 + d_2 \cdot 0 < 0$$

which implies $(0, \ker f_2)$ can only be a trivial subrepresentation, that is, $\ker f_2 = 0$.

Injectivity of f_2 implies $V_2 \subset V_1$ and $f_1 \circ f_2 = 0$ implies $V_2 \subset \ker(f_1)$. Set $t = f_2 \circ f_1 \in \mathrm{End}(V_1)$. Then $\mathrm{im}\ t \subset V_2 \subset \ker t$. Thus, we can rewrite the quiver variety as

$$\mathfrak{M}(d_2, d_1) = \{(V_2, t) \big| V_2 \subset V_1, \dim V_i = d_i, t \in \mathrm{End}(V_1), \mathrm{im}\ t \subset V_2 \subset \ker t\}$$

Now, we replace d_2, d_1 by w, d and V_1, V_2 by W, D for convenience. Define

$$\mathfrak{M}(w, d) = \{W, t) \big| W \subset D, \dim W = w, \dim D = d, t \in \mathrm{End}(D), \mathrm{im}\ t \subset W \subset \ker t\}$$

$$\mathfrak{M}(d) := \cup_w \mathfrak{M}(w, d) = \{(W, t) \big| W \subset D, t \in \mathrm{End}(D), \mathrm{im}\ t \subset W \subset \ker t\}$$

$$\mathfrak{M}(w, w+1, d) := \{(U, W, t) \big| t \in \mathrm{End}(D), \mathrm{im}\ t \subset U \subset W \subset \ker t, \dim U = w, \dim W = w+1\}$$

Then we can define

$$\begin{array}{ccccc} \mathrm{M}(d) & \xleftarrow[\pi_1]{} & \cup_w \mathfrak{M}(w, w+1, d) & \xrightarrow{\pi_2} & \mathfrak{M}(d) \\ (U, t) & \longleftarrow\!\shortmid & (U, W, t) & \longmapsto & (W, t) \end{array}$$

Example 4.1. We just consider some simple cases related to $\mathfrak{M}$.

When $d = 0$, $\mathfrak{M}(0) = \text{pt}$. When $d = 1$, either $\dim W = 1$, then $W = D$ or $\dim W = 0$, then $W = 0$. We also get $\mathfrak{M}(d) = \text{pt}$.

When $w = 1, d = 2$,

$$\begin{aligned}\mathfrak{M}(1,2) &= \{[w] \subset \mathbb{C}^2, t \in \mathrm{End}(D), \mathrm{im}\ t \subset [w] \subset \ker t\} \\ &= \{([w], t) : \mathrm{im}\ t \subset [w], t|_{[w]} = 0\} \\ &= T^*\mathbb{P}^1\end{aligned}$$

Moreover, it is easy to observe that $\mathfrak{M}(1,2,2) = \mathbb{CP}^1$.

What's π_1, π_2 explicitly? Examples?

Let $\pi_! : A \to B$ be a map between algebraic varieties A, B. Then we define

$$\pi_!(f)(x) = \sum_{y \in \pi^{-1}(x)} f(y)$$

a map between the abelian groups of functions on A and B. For example, for $Y \subset B$, we have $\pi_!(\mathbf{1}_Y) = \chi_q(\pi^{-1}(x) \cap Y) := |\pi^{-1}(x) \cap Y|$, the number of points in $\pi^{-1}(x) \cap Y$.

With notations above, we can now define the actions of $\mathbf{U}_q(\mathfrak{sl}_2)$ over the set of functions on $\mathfrak{M}(d)$ by

$$\begin{aligned}Ef(\cdot) &= q^{-\dim(\pi_1^{-1}(\cdot))}(\pi_1)_!\pi_2^* f(\cdot) \\ Ff(\cdot) &= q^{-\dim(\pi_2^{-1}(\cdot))}(\pi_2)_!\pi_1^* f(\cdot) \\ K^{\pm}f(\cdot) &= q^{\pm(d-2\dim(\cdot))} f(\cdot)\end{aligned}$$

Now, we give more definitions:

$$\mathfrak{M}^r(d) := \{(W,t) \in \mathfrak{M}(d) \big| \mathrm{rank}\ t = r\};\ \mathfrak{M}^r(w,d) := \{(W,t) \in \mathfrak{M}(w,d) \big| \mathrm{rank}\ t = r\}$$

Let

$$\mathcal{M}^r(w,d) = \mathbb{C}(q)\mathbf{1}_{\mathfrak{M}^r(w,d)};\ \mathcal{M}^r(d) = \oplus_w \mathcal{M}^r(w,d)$$

Proposition 4.2. *The action of $\mathbf{U}_q(\mathfrak{sl}_2)$ defined above endows $\mathcal{M}^r(d)$ with the structure of a $\mathbf{U}_q(\mathfrak{sl}_2)$-module and the map $\mathbf{1}_{\mathfrak{M}^r(w,d)} \mapsto v_{d-2w}$ (extended linearly) is an isomorphism*

$$\mathcal{M}^r(d) \cong V_{d-r}$$

of $\mathbf{U}_q$-modules. See 10.3.

Before proving this proposition, we need some lemmas:

Lemma 4.3. *For $W \subset D$,*

$$\{U \big| W \subset U \subset D, \dim U = u\} \cong \mathrm{Gr}_{u-\dim W}^{\dim D - \dim W}$$

Proof. Note that we have the following one-to-one correspondence

$$\{U \big| W \subset U \subset D, \dim U = u\} \xrightarrow{1-1} \{\bar{U} \big| \bar{U} \subset D/W, \dim \bar{U} = u - \dim W\}$$

□

Lemma 4.4. $\chi_q(\mathbb{P}^n) = \sum_{i=0}^n q^{2i}$.

Proof of 4.2. Take $(W,t) \in \mathfrak{M}^r(w,d)$. Then

$$E\mathbf{1}_{\mathfrak{M}^r(w+1,d)}(W,t) = q^{-\dim(\pi_1^{-1}(W,t))}(\pi_1)_!\pi_2^*\mathbf{1}_{\mathfrak{M}^r(w+1,d)}(W,t)$$

Then $\pi_1^{-1}(W,t) = \{U \big| W \subset U \subset \ker t, \dim U = w+1\}$. Since t is of rank r, $\ker t = d-r$ and so

$$\pi_1^{-1}(W,t) = \{U \big| W \subset U \subset \ker t, \dim U = w+1\} = \mathrm{Gr}_1^{d-r-w}$$

By definition, $\pi_2^*\mathbf{1}_{\mathfrak{M}^r(w+1,d)} = \mathbf{1}_{\mathfrak{M}^r(w,w+1,d)}$. So $(\pi_1)_!\mathbf{1}_{\mathfrak{M}^r(w-1,w,d)}(W,t) = \chi_q(\pi_1^{-1}(W,t) \cap \mathfrak{M}^r(w,w+1,d))$. We know that

$$\pi_1^{-1}(W,t) \cap \mathfrak{M}^r(w,w+1,d) = \mathrm{Gr}_1^{d-r-w}$$

Thus,

$$E\mathbf{1}_{\mathfrak{M}^r(w+1,d)}(W,t) = q^{-(d-r-w)} \sum_{i=0}^{2(d-r-w)} = [d-r-w]_q$$

Observe that when $(W,t) \notin \mathfrak{M}^r(w,d)$, $E\mathbf{1}_{\mathfrak{M}^r(w+1,d)}(W,t) = 0$. Thus, we get

$$E\mathbf{1}_{\mathfrak{M}^r(w+1,d)} = [d-r-w]_q\mathbf{1}_{\mathfrak{M}^r(w,d)}$$

Next, we compute $F\mathbf{1}_{\mathfrak{M}^r(w-1,d)}$. Recall that

$$F\mathbf{1}_{\mathfrak{M}^r(w-1,d)} = q^{-\dim(\pi_2^{-1}(W,t))}(\pi_2)_!\pi_1^*\mathbf{1}_{\mathfrak{M}^r(w-1,d)}(W,t)$$

Note that $\pi_1^*\mathbf{1}_{\mathfrak{M}^r(w-1,d)}(W,t) = \mathbf{1}_{\mathfrak{M}^r(w-1,w,d)}$. Thus, $F\mathbf{1}_{\mathfrak{M}^r(w-1,d)}(W,t) \neq 0$ if and only if $(W,t) \in \mathfrak{M}^r(w,d)$. Note that

$$\pi_2^{-1}(W,t) = \{U \big| \mathrm{im}(t) \subset U \subset W\} = \mathrm{Gr}_{w-1-r}^{w-r}, \quad \pi_2^{-1}(W,t) \cap \mathfrak{M}^r(w-1,w,d) = \mathrm{Gr}_{w-1-r}^{w-r}$$

and we have $\mathrm{Gr}_{w-1-r}^{w-r} \cong \mathbb{P}^{w-r}$. Thus, we get

$$F\mathbf{1}_{\mathfrak{M}^r(w-1,d)} = [w-r]_q\mathbf{1}_{\mathfrak{M}^r(w,d)}$$

Finally, it is easy to see that $K\mathbf{1}_{\mathfrak{M}^r(w,d)} = q^{d-2w}\mathbf{1}_{\mathfrak{M}^r(w,d)}$. So $\mathfrak{M}^r(d) \cong V_{d-r}$ as $\mathbf{U}_q(\mathfrak{sl}_2)$-modules. □

Let $\mathfrak{g}$ be a complex semi-simple algebra and Φ be the root system of $\mathfrak{g}$ with respect to a fixed Cartan subalgebra $\mathfrak{h}$. Let W be the Weyl group of Φ.

If Φ is irreducible, then there is a unique W-invariant scalar product $(-,-)$ on the root space $\mathfrak{h}^*$ generated by Φ such that $(\alpha,\alpha) = 2$ for all short roots $\alpha \in \Phi$. In fact, this scalar product is the dual of killing form over $\mathfrak{g}$.

Then we choose a simple root system $\Pi \subset \Phi$. Suppose $\Pi = \{\alpha_1, \alpha_2, ..., \alpha_n\}$. Define the **Cartan matrix** $A = (A_{ij})$ by setting

$$A_{ij} = \frac{2(\alpha_i,\alpha_j)}{(\alpha_j,\alpha_j)}$$

Example 4.5. Consider the Lie algebra

$$\mathfrak{sl}_3 = \{M \in \mathrm{M}_{3\times 3}(\mathbb{C}) \big| \mathrm{tr}(M) = 0\}$$

Then the Cartan algebra of $\mathfrak{sl}_3$ is given by

$$\mathfrak{h} = \left\{ \begin{pmatrix} h_1 & 0 & 0 \\ 0 & h_2 & 0 \\ 0 & 0 & h_3 \end{pmatrix} : h_1 + h_2 + h_3 = 0 \right\}$$

Then we can get root of $\mathfrak{sl}_3$:

$$\alpha_{ij} = \epsilon_i - \epsilon_j : \mathfrak{h} \to \mathbb{C}; \quad \mathrm{diag}(h_1, h_2, h_3) \mapsto h_i - h_j$$

for i, j and $\mathfrak{sl}_3 = \mathfrak{h} \oplus \mathfrak{g}_{\alpha_{ij}}$. Using the killing form on $\mathfrak{sl}_3$, we have $(\epsilon_i, \epsilon_j) = \delta_{ij}$. Let $\Pi = \{\alpha_{12}, \alpha_{23}\}$, then the weight space Γ is an abelian group generated by Π and

$$(\alpha_{12}, \alpha_{12}) = 2; (\alpha_{12}, \alpha_{23}) = -1; (\alpha_{23}, \alpha_{23}) = 2; (\alpha_{23}, \alpha_{12}) = -1.$$

Then the Cartan matrix is given by

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

We can write down generators and relations for $\mathfrak{g}$ using the Cartan matrix: The generators are given by $\{e_\alpha, f_\alpha, h_\alpha \big| \alpha \in \Pi\}$ and the relations are given by

$$[h_\alpha, h_\beta] = 0, \quad [e_\alpha, f_\beta] = \delta_{\alpha,\beta} h_\alpha, \quad [h_\alpha, e_\beta] = A_{\alpha\beta} e_\beta, \quad [h_\alpha, f_\beta] = -A_{\alpha\beta} f_\beta$$

and for all $\alpha \neq \beta$:

$$(\mathrm{ad}\ e_\alpha)^{1-A_{\alpha\beta}} e_\beta = 0, \quad (\mathrm{ad}\ f_\alpha)^{1-A_{\alpha\beta}} f_\beta = 0$$

The last two relations are called **Serre relations**.

Example 4.6. We can check that $\mathfrak{sl}_3$ is generated by $e_1 := E_{12}$, $e_2 := E_{23}$, $f_1 := E_{21}$, $f_2 := E_{32}$, $h_1 := E_{11} - E_{22}$ and $h_2 := E_{22} - E_{33}$. We can easily check that $\{e_\alpha, f_\alpha, h_\alpha \big| \alpha = 1, 2\}$ satifies the relations above.

Example 4.7. For $\mathfrak{sl}_3$, we denote it simple roots by $\{\alpha_1 := \alpha_{12}, \alpha_2 := \alpha_{23}\}$. Then $(\alpha_1, \alpha_2) = -1$ and $(\alpha_i, \alpha_i) = 2$ for $i = 1, 2$. Then we can write $(R3)$ as

$$K_\alpha E_\beta = q^{-1} E_\beta K_\alpha, \text{ if } \alpha \neq \beta; \quad K_\alpha E_\alpha = q^2 E_\alpha K_\alpha$$

and $(R5), (R6)$ are given by

$$E_1^2 E_2 - \binom{2}{1}_q E_1 E_2 E_1 + E_2 E_1^2 = 0; \quad E_2 E_1^2 - \binom{2}{1}_q E_2 E_1 E_2 + E_1 E_2^2 = 0$$

Note that

$$\binom{2}{1}_q = q + q^{-1}.$$

So we get

$$E_1^2 E_2 - (q + q^{-1}) E_1 E_2 E_1 + E_2 E_1^2 = 0; \quad E_2 E_1^2 - (q + q^{-1}) E_2 E_1 E_2 + E_1 E_2^2 = 0$$

Keeping everything the same as above, we have:

Definition 4.8. The quantized enveloping algebra $\mathbf{U}_q(\mathfrak{g})$ is defined as a $\mathbb{Q}(q)$-algebra with generators $\{E_\alpha, F_\alpha, H_\alpha \big| \alpha \in \Pi\}$ and relations:

$$(R1).\ K_\alpha K_\alpha^{-1} = 1 = K_\alpha^{-1} K_\alpha, \quad K_\alpha K_\beta = K_{\alpha+\beta};$$

$$(R2).\ K_a E_\beta = q^{(\alpha,\beta)} E_\beta K_\alpha;$$
$$(R3).\ K_\alpha F_\beta = q^{-(\alpha,\beta)} F_\beta K_\alpha;$$
$$(R4).\ E_\alpha F_\beta - F_\beta E_\alpha = \delta_{\alpha,\beta} \frac{K_\alpha - K_\alpha^{-1}}{q - q^{-1}}$$

and for $\alpha \neq \beta$:

$$(R5).\ \sum_{s=0}^{1-a_{\alpha\beta}} (-1)^s \binom{1-a_{\alpha\beta}}{s}_q E_\alpha^{1-a_{\alpha\beta}-s} E_\beta E_\alpha^s = 0$$
$$(R6).\ \sum_{s=0}^{1-a_{\alpha\beta}} (-1)^s \binom{1-a_{\alpha\beta}}{s}_q F_\alpha^{1-a_{\alpha\beta}-s} F_\beta F_\alpha^s = 0$$

We call $(R5), (R6)$ the **quantum Serre relations**.

Example 4.9. For $\mathfrak{sl}_2$, we know that the Cartan matrix is just $A = (2)$ and $(\alpha, \alpha) = 1$. Hence $(R5), (R6)$ are satisfies automatically. Because $(\alpha, \alpha) = 1$, $(R1) \sim (R4)$ is just as what we have defined before.

Definition 4.10. Let $\mathbf{U}_q(\mathfrak{n}^+)$ be the subalgebra of $\mathbf{U}_q(\mathfrak{g})$ generated by all E_α with $\alpha \in \Pi$ with relations $(R5)$.

Definition 4.11. Let $\mathbf{U}_q(\mathfrak{b}^+)$ be the subalgebra of $\mathbf{U}_q(\mathfrak{g})$ generated by E, K, K^{-1} with relations $(R1), (R2), (R5)$.

5 Quiver Representations

5.1 Quiver

Now, we give the formal definition of a quiver.

Definition 5.1. A quiver $\vec{Q}$ is a quadruple (I, H, t, s) with I vertex set, H edge set and $t, s : H \to I$ the target and sources. Most of time, we simply denote $\vec{Q}$ by (I, H).

Remark. We allow $\bar{Q}$ to have multiple edges and cycles, but no loops.

Example 5.2. Here are some examples of quiver, which we will encounter frequently:

(i) A_n: $A_n = (I, H)$ with $I = \{1, 2, \cdots, n\}$ and $H == \{h_i : i \to i+1\}_{i=1,\cdots n-1}$. The graph of A_n is:

$$1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \dots \longrightarrow n$$

(ii) K_n: $K_n = (I, H)$ with $I = \{1, 2\}$, $H = \{h_i : 1 \to 2\}_{i=1,\cdots,n}$. For examples, the graph of K_2 is

$$1 \rightrightarrows 2$$

and the graph of $K3$ is

$$1 \substack{\longrightarrow \\ \longrightarrow \\ \longrightarrow} 2$$

Definition 5.3. A representation of a quiver $\vec{Q}$ over a field k is a tuple $(V, \underline{x})$ with

$$V = \oplus_{i \in I} V_i; \quad \underline{x} = (x_h)_{h \in H} \in \bigoplus_{h \in H} \mathrm{Hom}(V_{s(h)}, V_{t(h)})$$

We say a representation of $\vec{Q}$ is nilpotent if there exists $N \gg 0$ such that for any $n > N$, $x_{h_n} \circ x_{h_{n-1}} \circ \cdots \circ x_{h_1} = 0$ for any path $h_1 h_2 \cdots h_n$ of length n in $\vec{Q}$. Denote the category of nilpotent representations by $\mathrm{Rep}_k^{nil}(\vec{Q})$.

Remark. When $\vec{Q}$ has no oriented edges, any representation is nilpotent.

Definition 5.4. We say $(U, \underline{y})$ is a subrepresentation of $(V, \underline{x})$ if $U_i \subseteq V_i$ for all $i \in I$ and $y_h = x_h|_{U_{s(h)}}$.

Here we just state and prove some basic properties of $\mathrm{Rep}_k^{nil}(\vec{Q})$.

Proposition 5.5. $\mathrm{Rep}_k^{nil}(\vec{Q})$ *is an abelian, k-linear category satisfying Krull-Schmidt property.*

$\mathrm{Rep}_k^{nil}(\vec{Q})$ is abelian so kernels, cokernels in $\mathrm{Rep}_k^{nil}(\vec{Q})$ exist. Krull-Schmidt properties implies that every object in $\mathrm{Rep}_k^{nil}(\vec{Q})$ decomposes as a finite direct sum of indecomposables.

Example 5.6. Let $\vec{Q}$ be a quiver. Let $i \in I$ and $S_i = (V, \underline{x})$ such that

$$V_i = k, \quad V_j = 0, j \neq i, \quad \underline{x} = 0$$

which is by definition a representation of $\vec{Q}$.

Moreover, $\{S_i\}_{i \in I}$ plays an important role in $\mathrm{Rep}_k^{nil}(\vec{Q})$:

Proposition 5.7. *The collection* $\{S_i\}_{i \in I}$ *is a complete set of* ***simple*** *objects of* $\mathrm{Rep}_k^{nil}(\vec{Q})$.

Proof. Let $(V, \underline{x})$ be a simple representation of $\vec{Q}$. Since $(V, \underline{x}) \in \mathrm{Rep}_k^{nil}(\vec{Q})$, there exists a path $h_1 \cdots h_i$ such that

$$x_{h_i} \circ x_{h_{i-1}} \circ \cdots \circ x_{h_1} \neq 0; \quad x_{h_{i+1}} \circ x_{h_i} \circ \cdots \circ x_{h_1} = 0$$

for any $h_{i+1} \in H$ with $s(h_{i+1}) = t(h_i)$. Then take $v \in V_{s(h_1)}$ such that $u := x_{h_i} \circ x_{h_{i-1}} \circ \cdots \circ x_{h_1}(v) \neq 0$. Consider $(U, \underline{y})$

$$U_{t(h_i)} = \mathrm{span}_k\{u\}, \quad U_j = 0, j \neq t(h_i), \quad \underline{y} = 0$$

Then $(U, \underline{y})$ is a subrepresentation of $(V, \underline{x})$ and it is isomorphic to $S_{t(h_i)}$. For $(V, \underline{x})$ is simple, $(V, \underline{x}) \cong S_{t(h_i)}$. □

Corollary 5.8. $\mathrm{Rep}_k^{nil}(\vec{Q})$ *is of finite length and*

(i). Any representation $(V, \underline{x})$ *in* $\mathrm{Rep}_k^{nil}(\vec{Q})$ *admits a finite map series consisting of* S_i*'s;*

(ii). $K(\mathrm{Rep}_k^{nil}(\vec{Q})) \cong \mathbb{Z}^{\oplus I}$.

Let $c_{ij} := \#\{h \in H \big| s(h) = i, t(h) = j\}$. Then we can directly compute the Euler form of two simple objects in $\mathrm{Rep}_k^{nil}(\vec{Q})$:

Proposition 5.9. *Let* S_i, S_j *be as above and* $k = \mathbb{F}_q$. *Then*

$$\langle S_i, S_j \rangle_a = \delta_{ij} - c_{ij}, \quad \langle S_i, S_j \rangle_m = q^{\frac{1}{2}(\delta_{ij} - c_{ij})}$$

Proof. We use the fact that $\dim \mathrm{Ext}^1(S_i, S_j) = c_{ij}$. By Yoneda's interpretation for extensions [Exercise III.6.1 [Har77]], $\mathrm{Ext}^1(S_i, S_j)$ corresponds to the exacts sequence

$$0 \longrightarrow S_j \longrightarrow E \longrightarrow S_i \longrightarrow 0$$

up to isomorphism.

By the short exact sequence above, we get $E_i = k$, $E_j = k$ and $E_v \neq 0$ when $v \neq i, j$. For any $a : i \to j$, $E_a : k \longrightarrow k$ is given by $\lambda_a \in k$. Since E contains S_j as a subrepresentation, for any $b : j \longrightarrow i$, $E_b = 0$. Hence, we conclude that $\dim \mathrm{Ext}^1(S_i, S_j) = c_{ij}$. □

Now, fix a quiver $\vec{Q}$. Let $Q = \bigoplus_{i \in I} \mathbb{Z}\alpha_i$ be the root lattice. Let $(,)$ be the restriction of the Cartan killing form for Q given by

$$(u, v) = \sum_{i \in I} u_i v_i - \sum_{\alpha : i \to j} \alpha_i \beta_j$$

with $u = (u_i)_{i \in I}$ and $v = (v_i)_{i \in I}$.

Recall that we can define symmetry Euler form over $K(\mathrm{Rep}^{nil}(\vec{Q}))$ via

$$\langle [M], [N] \rangle := \dim_k \mathrm{Hom}(M, N) - \dim_k \mathrm{Ext}^1(M, N)$$

By $[M] = \oplus_{i \in I}[m_i S_i]$, we get

Corollary 5.10. *The map*

$$\begin{aligned} \rho : K(\mathrm{Rep}^{nil}(\vec{Q})) &\longrightarrow Q \\ \bar{V} &\longmapsto \sum_{i \in I} (\dim V_i)\alpha_i \end{aligned}$$

is an isomorphism of $\mathbb{Z}$*-modules, maps the symmetry Euler form* $\langle\ ,\ \rangle$ *to the Cartan killing form* $(\ ,\ )$.

Definition 5.11. We say a quiver $\vec{Q}$ is

(i) **of finite type** if $\mathrm{Rep}^{nil}(\vec{Q})$ has only finitely many indecomposables;

(ii) **tame** if the indecomposable lying in each class in $K(\mathrm{Rep}^{nil}(\vec{Q}))$ can be arranged in finitely many one-parameter family;

(iii) **wild** if it is neither of finite type nor tame.

Example 5.12. A_2 is a quiver of finite type for $\mathrm{Rep}^{nil}(A_2)$ has only finitely many indecomposables, say M_{12}, S_1, S_2 (We will show this later!).

K_2 is a tame quiver. In fact, consider indecomposable representations of K_2 with $V_1 = k, V_2 = k$:

$$x_{h_1} = x_{h_2} = 0; \quad x_{h_1} = 0, x_{h_2} = 1; \quad x_{h_1} = 1, x_{h_2} = 0; \quad x_{h_1} = 1, x_{h_2} = \lambda \in k^*$$

We can see that the last two cases are parameterized by $\lambda \in k$.

5.2 Representations of Quivers of Finite Type

Now, let's introduce some results in the representations of quivers. First of all, we need to give a more general concept of Cartan matrix:

Definition 5.13. We say $A = (a_{ij})$ is a Cartan matrix if

$$a_{ii} = 2; \quad a_{ij} \leq 0 \text{ if i} \neq j; \quad a_{ij} = 0 \Longrightarrow a_{ji} = 0.$$

Definition 5.14. A **Dynkin diagram** of A is a graph Ω consisting n vertices and p_i and p_j are joint by $k_{ij} = \min\{|a_{ij}|, |a_{ji}|\}$ edges.

Given a Cartan matrix A, we can assign it a Lie algebra via 4, called **Kac-Moody algebra**.

Example 5.15. For A_2, its Cartan matrix is given by

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$$

which is just the Cartan algebra of $\mathfrak{sl}_3$.

For K_2, its Cartan matrix is

$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$$

and the Dykin diagram corresponding is

$$1 \;\overline{\overline{\quad}}\; 2$$

Obviously, the **Dykin diagram** of a quiver $\vec{Q}$ is just the underlying graph of $\vec{Q}$.

Now, given a quiver $\vec{Q}$, we can define a matrix $A = (a_{ij})$ such that

$$a_{ij} = 2\delta_{ij} - c_{ij} - c_{ji}$$

which is a Cartan matrix and hence, we can associate it with a Lie algebra $\mathfrak{g}_{\vec{Q}}$, via 4. Here are some results about the Cartan matrix and the associated Lie algebra $\mathfrak{g}$:

Theorem 5.16 (Cartan)**.** *An irreducible matrix $A = (a_{ij})$ with integer coefficient is the Cartan matrix of a Lie algebra if and only if*

$$a_{ii} = 2; \quad a_{ij} \leq 0 \text{ if } i \neq j; \quad a_{ij} = 0 \Longrightarrow a_{ji} = 0.$$

and A is positive definite.

Theorem 5.17. *(Cartan) The set of Cartan matrices of* ***simple*** *Lie algebra corresponds to the Dynkin diagram:*

$$A_r, B_r, C_r, D_r, E_6, E_7, E_9, F_4, G_2$$

Moreover, a ***symmetry*** *Cartan matrix is positive definite if and only if the Dynkin diagram is of type A, D, E.*

We can use the Cartan matrix of a quiver to see whether or not it is of finite type:

Theorem 5.18. *(Gabriel) The category $\mathrm{Rep}^{nil}(\vec{Q})$ is of finite type if and only if $\mathfrak{g}_{\vec{Q}}$ is a simple Lie algebra, that is, if and only if $\mathfrak{g}_{\vec{Q}}$ is a simple Lie algebra.*

Moreover, $(V, \underline{x}) \mapsto \dim(V_i)\alpha_i \in Q$ is a bijection between the set of indecomposable objects and the set Δ^+ of ***positive roots****.*

For any quiver $\vec{Q}$, the associated Cartan matrix is always symmetry. Combining with 5.17, we have the following result:

Corollary 5.19. *$\vec{Q}$ is of finite type if and only if the underlying graph is of type A, D, E.*

Remark. By simple computation, the underlying graph of K_2 is just B_2, to which there exists a simple Lie algebra corresponding. However, K_2 is not a quiver of finite type, for the Cartan matrix of K_2 is not positive definite. This is because for the Lie algebra $\mathfrak{g}$ corresponds to B_2, its Cartan matrix is **Not** symmetry.

Now, we can describe the indecomposables in $\mathrm{Rep}^{nil}(\vec{Q})$ with $\vec{Q}$ of finite type:

Example 5.20 (Indecomposables of A_n)**.** Consider A_n with the associated Lie algebra $\mathfrak{sl}_n$. If $\{\alpha_1, \cdots, \alpha_n\}$ are simple roots, then

$$\Delta^+\{\alpha_i + \alpha_{i+1} + \cdots + \alpha_j \big| i \leq j\}$$

The indecomposable $I_{i\cdots j}$ corresponds to $\alpha_i + \alpha_{i+1} + \cdots + \alpha_j$ is

$$\begin{array}{ccccccccccccccc}
0 & & 0 & & \cdots & & k & & k & & \cdots & & k & & \cdots & & 0 \\
\bullet & \dashrightarrow & \bullet & \dashrightarrow & \cdots & \dashrightarrow & \bullet & \xrightarrow{\cong} & \bullet & \xrightarrow{\cong} & \cdots & \xrightarrow{\cong} & \bullet & \dashrightarrow & \cdots & \dashrightarrow & \bullet \\
1 & & 2 & & \cdots & & i & & i+1 & & \cdots & & j & & \cdots & & n
\end{array}$$

Moreover, we can directly compute that

$$\mathrm{Hom}(I_{i\cdots j}, I_{i'\cdots j'}) = \begin{cases} k, & \text{if } i' \leq i \leq j \leq j'; \\ 0, & \text{otherwise.} \end{cases}$$

Example 5.21. Suppose $\vec{Q}$ is

$$\begin{tikzcd}
 & \bullet^{3} \arrow[d] & \\
\bullet_{1} \arrow[r] & \bullet_{2} & \bullet_{4} \arrow[l]
\end{tikzcd}$$

The associated Lie algebra is $\mathfrak{so}_8(\mathbb{C})$. Recall that the root system of $\mathfrak{so}_8$ is

$$\Delta = \{\varepsilon_i - \varepsilon_j, \varepsilon_i + \varepsilon_j, -\varepsilon_i - \varepsilon_j, -\varepsilon_i + \varepsilon_j \big| 1 \leq i < j \leq 4\}$$

We can set

$$\alpha_1 = \varepsilon_1 - \varepsilon_2, \quad \alpha_2 = \varepsilon_2 - \varepsilon_3, \quad \alpha_3 = \varepsilon_3 - \varepsilon_4, \quad \alpha_4 = \varepsilon_3 + \varepsilon_4.$$

Then we can get a positive root system related to the simple roots $\alpha_1, \alpha_2, \alpha_3, \alpha_4$:

$$\Delta^+ = \{\alpha_1, \alpha_2, \alpha_3, \alpha_4\} \cup \{\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_2 + \alpha_4\}$$

$$\cup \{\alpha_1+\alpha_2+\alpha_3, \alpha_1+\alpha_2+\alpha_4, \alpha_2+\alpha_3+\alpha_4\}$$
$$\cup \{\alpha_1+\alpha_2+\alpha_3+\alpha_4\} \cup \{\alpha_1+2\alpha_2+\alpha_3+\alpha_4\}$$

By Gabriel's theorem 5.18, the indecomposables of $\mathrm{Rep}^{nil}(\vec{Q})$ are

$$I_2, I_{12}, I_{23}, I_{24}, I_{123}, I_{124}, I_{234}, I_{1234}, I_{12^234}$$

For example, the indecomposable corresponding to I_{12^234} is

$$\begin{tikzcd} & k \arrow[d] & \\ k \arrow[r] & k^2 & k \arrow[l] \end{tikzcd}$$

via $V \mapsto \sum_{i=1}^4 (\dim V_i)\alpha_i = \alpha_1 + 2\alpha_2 + \alpha_3 + \alpha_4$.

Now, come back to the Hall algebra of $\mathrm{Rep}_k^{nil}(\vec{Q})$ with $k = \mathbb{F}_q$.

Example 5.22. As we have assumed above, $\vec{Q}$ is a quiver without loops. Thus, for any S_i, by 5.9,

$$\mathrm{Ext}^1(S_i, S_i) = 0$$

Thus, we can show that

$$[S_i]\cdot[S_i] = v\#\{N \subset S_i \oplus S_i \big| N \cong S_i, S_i \oplus S_i/N = S_i\} = v(v^2+1)[S_i^{\oplus 2}]$$

Moreover, we can prove by induction

$$[S_i]^l = [l]_q! v^{\frac{l(l-1)}{2}}[S_i^{\oplus 2}]$$

Suppose that $[S_i]^{l-1} = [l-1]_q! v^{\frac{(l-1)(l-2)}{2}}[S^{\oplus l-1}]$. Then

$$[S_i^{\oplus l-1}]\cdot[S_i] = v^{l-1}\#\{N \subset S_i^{\oplus l} \big| N \cong S_i, S^{\oplus l}/N \cong S_i^{\oplus l-1}\}[S_i^{\oplus l}] = v^{l-1}\#|\mathbb{P}^l(\mathbb{F}_q)|[S_i^{\oplus l}]$$

We know that

$$\#|\mathbb{P}^l(\mathbb{F}_q)| = \binom{l}{1}_q = \frac{q^l-1}{q-1}$$

By the definition of $[l]_q!$, we get the result.

Via denoting

$$[l]_v! = \prod_{i=1}^{l} \frac{v^i - v^{-i}}{v - v^{-1}}$$

we can rewrite our result as

$$[S_i^l] = v^{l(l-1)}[l]_v![S_i^{\oplus 2}]$$

which will help us in some further theorem.

Example 5.23. Keep the notations in 5.20. Recall that in 1.10, we have shown

$$[S_1]\cdot[S_2] = v^{-1}([S_1 \oplus S_2] + [I_{12}]); \quad [S_2]\cdot[S_1] = [S_1 \oplus S_2]$$

with $v = q^{\frac{1}{2}}$.

Let us compute $[S_2]\cdot[S_1]^2, [S_1]^2\cdot[S_2]$, $[S_1]\cdot[S_2]\cdot[S_1]$ and find relations between them.

$$[S_1]^2\cdot[S_2] = v^{-1}([S_1]\cdot[S_1] \oplus [S_2] + [S_1]\cdot[I_{12}])$$

By definition:

$$\begin{aligned}[S_1]\cdot[S_1]\oplus[S_2] &= \langle[S_1],[S_1]\oplus[S_2]\rangle_m \sum_{[R]} \#\{N\subset R \big| N\cong S_1\oplus S_2, R/N\cong S_1\}[R] \\ &= [S_1\oplus I_{12}] + (v^2+1)[S_1^{\oplus 2}\oplus S_2]\end{aligned}$$

since 1). $\mathrm{Hom}(S_1, S_1\oplus S_2) = k$ and $\mathrm{Ext}(S_1, S_1\oplus S_2) = \mathrm{Ext}^1(S_1, S_2) = k$; 2), in $S_1\oplus S_1\oplus S_2$, the $N\subset S_1\oplus S_1$ such that $N\cong S_1$ are just all lines in $S_1\oplus S_1$, whose number is $v^2+1 = \#\mathbb{P}(k)$.

$$[S_1]\cdot[I_{12}] = \langle[S_1],[I_{12}]\rangle_m \sum_{[R]} \#\{N\subset R \big| N\cong I_{12}, R/N\cong S_1\} = v^2[S_1\oplus I_{12}]$$

since: 1). By 5.18, all R are of the form $S_1^{\oplus n_1}\oplus S_2^{\oplus n_2}\oplus I_{12}^{\oplus n_{12}}$. So $R = S_1^{\oplus 2}\oplus S_2$ or $R = I_{12}\oplus S_1$. We can observe that the former one doesn't have any subspace that is isomorphic to I_{12}. 2). Consider $k\oplus k\longrightarrow k$ given by $(1,0)$. Then for any line L that is not contained in S_1, we have $I_{12}|_L = \mathrm{Id}_L$. The number of such lines are $v = \#\mathbb{P}(k)-1$. Thus, we get

$$[S_1]^2\cdot[S_2] = v^{-1}(v^2+1)([S_1^2\oplus S_2] + [I_{12}\oplus S_1])$$

Next, we compute $[S_2]\oplus[S_1]^2$: Note that $[S_1]^2 = v(v^2+1)[S_1^{\oplus}]$ where v comes form $\mathrm{Hom}(S_1,S_1) = k$ and $v^2+1 = \#\mathbb{P}(k)$. Then

$$\begin{aligned}[S_2]\cdot[S_1]^2 &= v(v^2+1)[S_2]\cdot[S_1\oplus S_1] \\ &= v(v^2+1)\sum_R \{N\subset R \big| N\cong S_1\oplus S_1, R/N\cong S_2\} \\ &= v(v^2+1)[S_1^{\oplus 2}\oplus S_2]\end{aligned}$$

Finally, we compute $[S_1]\cdot[S_2]\cdot[S_1]$.

$$\begin{aligned}[S_1]\cdot[S_2]\cdot[S_1] &= [S_1]\cdot[S_1\oplus S_2] \\ &= \langle S_1, S_1\oplus S_2\rangle_m \sum_R \#\{N\subset R \big| N\cong S_1\oplus S_2, R/N\cong S_1\} \\ &= (v^2+1)[S_1^{\oplus 2}\oplus S_2] + [I_{12}\oplus S_1]\end{aligned}$$

since $\mathrm{Hom}(S_1, S_1\oplus S_2) = k$ and $\mathrm{Ext}^1(S_1, S_1\oplus S_2) = k$.

Thus, we get (directly compute):

$$[S_1]^2\cdot[S_2] - (v+v^{-1})[S_1]\cdot[S_2]\cdot[S_1] + [S_2]\cdot[S_1]^2 = 0$$

Now, we compare the results in 4.7 with 5.23. Note that for $U_q(\mathfrak{sl}_3)$, the Serre relation is just

$$E_1^2E_2 - (q^{-1}+q)E_1E_2E_1 + E_2E_1^2 = 0$$

In fact, we have the following result:

Theorem 5.24 (Ringel-Green)**.** *The assignment $E_i\mapsto[S_i]$, $K_i\mapsto \mathbf{k}_{S_i}$ for $i\in I$ extends to an embedding of Hopf algebras*

$$\Psi : \mathbf{U}_v(\mathfrak{b}_+)\longrightarrow \tilde{\mathbf{H}}_{\vec{Q}}$$

The map Ψ is an isomorphism if and only if $\vec{Q}$ is of finite type, that is, $\vec{Q}$ is of type A, D, E.

Remark. For the first observation of this theorem, note that the graded structure on $U_q(\mathfrak{n}^+_{\vec{Q}})$ and the hall algebra (No extended !) $\mathbf{H}_{\vec{Q}}$.

For any $\mathfrak{g}_{\vec{Q}}$, $\mathfrak{g}_{\vec{Q}} = \mathfrak{n}^-_{\vec{Q}} \oplus \mathfrak{h} \oplus \mathfrak{n}^+_{\vec{Q}}$, where $\mathfrak{n}^+$ is generated by simple roots $\alpha_1, ..., \alpha_n$ and

$$\mathfrak{n}^+ = \bigoplus_{\alpha \in \Delta^+} \mathfrak{g}_\alpha$$

Hence, $U(\mathfrak{n}^+)$ is graded by $\mathbb{N}^n = \oplus_{i=1}^n \alpha_i \mathbb{N}$ and

$$\dim U(\mathrm{n}^+)_\alpha = \#\{\text{ways of combination of } \alpha_1, \cdots, \alpha_n \text{ to get } \alpha\}$$

Also, by 1, there is a graded structure on $\mathbf{H}_{\vec{Q}}$ given by the Grothendieck group and for $\alpha \in K(\vec{Q}) \cong \mathbb{Z}^I$

$$\dim(\mathbf{H}_{\vec{Q}})_\alpha \#\{\text{ways of combination of } [S_1], \cdots, [S_n] \text{ to get } \alpha\}$$

By 5.18, if $\vec{Q}$ is of finite type, then these two graded dimensions are equal.

6 Ringel-Green Theorem

Let $\mathfrak{g}$ be the Kac-Moody algebra associated to the Euler matrix $A = (a_{ij})_{i,j\in I}$ of $\vec{Q}$ and let $\mathbf{U}_v(\mathfrak{g}_{\vec{Q}})$ be the quantum group which is specialized at $q = v$.

Remark 6.1. There may be some confusion here. When $k = \mathbb{F}_q$, we have $v = \sqrt{q}$. However, in the quantum group $\mathbf{U}_q(\mathfrak{g})$, the parameter q is regarded as an indeterminate.

Let $\mathbf{U}_v(\mathfrak{b}_+)$ be the positive half of the quantum group $\mathbf{U}_v(\mathfrak{g})$ and let $U_v(\mathfrak{n}^+)$ be the quantum group generated by E_i.

We are going to prove:

Theorem 6.2 (Ringel-Green)**.** *The assignment $E_i \mapsto [S_i]$, $K_i \mapsto \mathbf{k}_{S_i}$ for $i \in I$ extends to an embedding of Hopf algebras*

$$\Psi : \mathbf{U}_v(\mathfrak{b}_+) \longrightarrow \tilde{\mathbf{H}}_{\vec{Q}}$$

The map Ψ is an isomorphism if and only if $\vec{Q}$ is of finite type, that is, $\vec{Q}$ is of type A, D, E.

Proof. **Step 1**: We need to verify that Ψ is well-defined. Recall that by the definition of $\mathbf{U}_v(\mathfrak{b}_+)$

$$\begin{aligned}
&(R1).\ K_iK_j = K_jK_i, \quad K_iE_jK_i^{-1} = v^{a_{ij}}E_j, \forall i,j \in I;\\
&(R5).\ \sum_{l=0}^{1-a_{ij}} (-1)^l \binom{1-a_{ij}}{l}_v E_i^l E_j E_i^{1-a_{ij}-l} = 0;\\
&(3).\ \Delta(K_i) = K_i \otimes K_i, \quad \Delta(E_i) = E_i \otimes 1 + K_i \otimes E_i;\\
&(4).\ S(K_i) = K_i^{-1}, \quad S(E_i) = -K_i^{-1}E_i.
\end{aligned}$$

Hence, we need to check $(R1), (R5), (3), (4)$ in $\tilde{\mathbf{H}}_{\vec{Q}}$.

Here by the definition of $\tilde{\mathbf{H}}_{\vec{Q}}$ 2, $(R1), (3), (4)$ are immediately. The most difficult part is to verify the Serre relations. To begin with, we set $r = c_{ij}$, $s = c_{ji}$ and $t = r + s = -a_{ij}$. Define

$$[S_i]^{(l)} := \frac{[S_i]^l}{[l]_v!} = v^{l(l-1)}[S_i^{\oplus l}]$$

Since $\langle S_i^{\oplus l}, S_j\rangle_m = v^{-lr}$,

$$[S_i]^{(l)} \cdot [S_j] = v^{l(l-1)-lr} \sum_{M\in I_1} [M]$$

where $I_1 = \{M \big| \exists N \subset M, N \cong S_j, M/N \cong S_i^{\oplus l}\}$.

For a representation L of $\vec{Q}$ of dim $(r+1)\varepsilon_i + \varepsilon_j$, we define

$$U_L = \bigcap_{h:i\to j} \ker x_h, \quad V_L = \sum_{h:j\to i} \operatorname{im} x_h$$

and set $u_L = \dim(U_L)$, $w_L = \dim(V_L)$. Then

Claim 6.3.

$$[S_i]^{(l)} \cdot [S_j] \cdot [S_i]^{(n)} = v^{-ns-lr+nl+l(l-1)+n(n-1)} \sum_{[M]} P_{M,n}[M]$$

where $P_{M,n} = 0$ unless $V_M \subset U_M$, in which case

$$P_{M,n} = \#\mathrm{Gr}(n - w_M, u_N - w_N) = v^{(u_M-n)(n-w_M)} \binom{u_M - w_M}{n - w_M}_v$$

Proof of the Claim.

$$\begin{aligned}[S_i]^{(l)}[S_j][S_i]^{(n)} &= v^{l(l-1)-lr}\cdot v^{n(n-1)}\sum_{M\in I_1}[M]\cdot[S_i^{\oplus n}]\\ &= v^{l(l-1)+n(n-1)-lr}\sum_{M\in I_1}\langle M,S_i^{\oplus n}\rangle_m\sum_N \#\{L\cong N\big|L\cong S_i^{\oplus n}, N/L\cong M\}[N]\end{aligned}$$

First, we compute $\langle M, S_i^{\oplus n}\rangle_m$. For M, we have a short exact sequence

$$0\longrightarrow S_j\longrightarrow M\longrightarrow S_i^{\oplus l}\longrightarrow 0$$

Then applying $\mathrm{Hom}(-,S_i^{\oplus l})$ on it, we get $\langle M, S_i^{\oplus n}\rangle_m = v^{ln-ns}$. Next, if $M\in I$, we can compute via

$$\begin{tikzcd}
 & i \arrow[d] \arrow[r, shift left, "0"] & j \arrow[l, shift left, "0"] \arrow[d, "l"] \\
S_j & & \\
M \arrow[d] & \bullet \arrow[r, shift left, "h_M"] & \bullet \arrow[l, shift left, "h'_M"] \arrow[d] \\
S_i^{\oplus l} & \bullet \arrow[r, shift left, "0"] & \bullet \arrow[l, shift left, "0"]
\end{tikzcd}$$

which implies $h'_M = 0$. Moreover, we know that $M_i = k^{\oplus l}$ and $M_j = k$. Next, we consider N such that $L\subset N$, $L\cong S_i^{\oplus n}$ and $N/L\cong M$.

$$\begin{tikzcd}
 & i \arrow[d, "i"] \arrow[r, shift left, "0"] & j \arrow[l, shift left, "0"] \arrow[d, "l"] \\
S_i^{\oplus n} & & \\
N \arrow[d, "p"] & \bullet \arrow[r, shift left, "h_N"] & \bullet \arrow[l, shift left, "h'_N"] \arrow[d] \\
M & \bullet \arrow[r, shift left, "h_M"] & \bullet \arrow[l, shift left, "0"]
\end{tikzcd}$$

By the commutativity of this diagram, $\mathrm{im}\, h'_N \subseteq \ker p = \mathrm{im}\, i$ and $\mathrm{im}\, i\subseteq \ker h_N$, where $\ker h_N, \mathrm{im}\, h'_N$ are just defined as U_N, V_N before. Fix such N. We want to choose L such that $U_N\subset L_i\subset V_N$. By 4.3, the number of such L are $\#\mathrm{Gr}(n-w_M, u_M-w_M)$. Note that once we choose such L, h_M is determined. Thus,

$$\begin{aligned}[S_i]^{(l)}[S_j][S_i]^{(n)} &= v^{l(l-1)+n(n-1)-lr}\sum_{M\in I_1}\langle M,S_i^{\oplus n}\rangle_m\sum_N \#\{L\cong N\big|L\cong S_i^{\oplus n}, N/L\cong M\}[N]\\ &= v^{l(l-1)+n(n-1)-lr+ln-ns}\sum_N P_{N,n}[N]\end{aligned}$$

with $P_{N,n}$ defined before. □

To verify $(R5)$, setting $n=t+1-l$, obtain

$$\sum_{l=0}^{t+1}(-1)^l[S_i]^{(l)}[S_j][S_i]^{(t+1-l)} = \sum_{[N],V_N\subset U_N} P_M[N]$$

where

$$P_N = \sum_{l=0}^{t+1}(-1)^l v^{-(t+1-l)s-lr+(t+1-l)l+l(l-1)+(t+1-l)(t+1-l-1)+(u_N-t-1+l)(t+1-l-w_N)}\binom{u_N-w_N}{t+1-l-w_N}_v$$

$$= v^{(-t+1)s-u_N w_N} \sum_{n=0}^{t+1} (-1)^{t+1-n} v^{-(2s+1-u_N-w_N)n} \binom{u_N - w_N}{n - w_N}_v$$

As we have seen $N_j = k$, so $s \geq w_M$. Observe that $u_N \geq s+1$. This is because U_M is the intersection of r linear spaces whose dimensions are at least t for $\dim N_i = t+1$.Then use

Lemma 6.4. *Let $m \geq 1$ and let $1-m \leq d \leq m-1$ with $d \equiv m-1 \mod 2$. Then*

$$\sum_{n=0}^{m} (-1)^n v^{-db} \binom{m}{n}_v = 0$$

We get

$$\sum_{l=0}^{t+1} (-1)^l [S_i]^{(l)} [S_j] [S_i]^{(t+1-l)} = \sum_{[N], V_N \subset U_N} P_M[N]$$

which finishes the proof of Step 1.

Step 2. We will show that Ψ is injective. Before proving this result. We need more tools, say Green's scalar product.

Proposition 6.5. *The non-degenerate scalar product $(,) : \mathbf{H}_{\mathcal{A}} \otimes \mathbf{H}_{\mathcal{A}} \longrightarrow \mathbb{C}$ defined by*

$$([M],[N]) := \frac{\delta_{M,N}}{a_M}$$

is a Hopf pairing, i.e. for any triple x, y, z of elements in $\mathbf{H}_{\mathcal{A}}$ we have $(xy, z) = (x \otimes y, \Delta(z))$.

Proof. Since the scalar product is linear, it is enough to check for $x = [M], y = [N], z = [R]$.

$$\begin{aligned}
(xy, z) &= ([M][N], [R]) \\
&= \langle M, N \rangle_m \#\{L \subset R \big| L \cong N, R/L \cong M\}[R] \frac{1}{a_R} \\
&= \langle M, N \rangle_m \frac{1}{a_M a_N a_R} \mathbf{P}^R_{M,N} \\
(x \otimes y, \Delta(z)) &= ([M] \otimes [N], \langle M, N \rangle_m \frac{1}{a_R} \mathbf{P}^R_{M,N} [M] \otimes [N]) \\
&= \langle M, N \rangle_m \frac{1}{a_M a_N a_R} \mathbf{P}^R_{M,N}
\end{aligned}$$

□

Corollary 6.6. *The scalar product $\tilde{\mathbf{H}}_{\mathcal{A}} \otimes \tilde{\mathbf{H}}_{\mathcal{A}} \longrightarrow \mathbb{C}$ defined by*

$$([M]\mathbf{k}_\alpha, [N]\mathbf{k}_\beta) := \frac{\delta_{M,N}}{a_M} (\alpha, \beta)_m$$

is a Hopf pairing, i.e., for any triple of elements in $\tilde{\mathbf{H}}_{\mathcal{A}}$, we have $(xy, z) = (x \otimes y, \Delta z)$.

Now, back to step 2, let $\{,\} : \mathbf{U}_v(\mathfrak{b}_+) \otimes \mathbf{U}_v(\mathfrak{b}_+) \longrightarrow \mathbb{C}$ be the pull-back of Green's scalar product under Φ on $\tilde{H}_{\vec{Q}}$. As Φ is a morphism of Hopf algebra (verified in Step 1!), $\{,\}$ is also a Hopf pairing on $\mathbf{U}_v(\mathfrak{b}_+)$ and

$$\{E_i, E_j\} = \frac{\delta_{ij}}{v^2 - 1}, \quad \{K_i, K_j\} = v^{a_{ij}}, \quad \{E_i, K_j\} = 0$$

In fact, we have the following result for quantum groups:

Theorem 6.7. *There exists a unique Hopf pairing on* $\mathbf{U}_v(\mathfrak{b}_+)$ *such that*

$$\{E_i, E_j\} = \frac{\delta_{ij}}{v^2-1}, \quad \{K_i, K_j\} = v^{a_{ij}}, \quad \{E_i, K_j\} = 0$$

It is non-degenerate.

$\{,\}$ on $\mathbf{U}_v(\mathfrak{b}_+)$ is a non-degenerate pairing. In particular, when restricting on $\mathbf{U}_v(\mathfrak{n}_+)$, we have

$$\ker \Phi \cap \mathbf{U}_v(\mathfrak{n}_+) = \{0\}$$

which implies $\Phi|_{\mathbf{U}_v(\mathfrak{n}_+)} : \mathbf{U}_v(\mathfrak{n}_+) \longrightarrow \mathbf{H}_{\vec{Q}}$ is injective. The injection on $\mathbf{U}_v(\mathfrak{b}_+)$ follows from $\mathbf{U}_v(\mathfrak{b}_+) = \mathbb{C}[K_i^{\pm}] \otimes \mathbf{H}_{\vec{Q}}$ and $\tilde{\mathbf{H}}_{\vec{Q}} = \mathbb{C}[k_{S_i}^{\pm}] \otimes \mathbf{H}_{\vec{Q}}$.

Step 3: Prove surjective when $\vec{Q}$ is of finite type. We will show this via observing that the graded dimension of $\mathbf{U}_v(\mathfrak{b}_+)$ and $\tilde{\mathbf{H}}_{\vec{Q}}$ agree.

Claim 6.8.

$$\dim(\mathbf{U}_v(\mathfrak{b}_+))_\beta = \#\{(n_\alpha) \in \mathbb{N}^{\Delta^+} \big| \sum n_\alpha \alpha = \beta\}$$

$$\dim(\tilde{\mathbf{H}}_{\vec{Q}})_\beta = \#\{(m_I) \in \mathbb{N}^{\mathrm{Indep}(\vec{Q})} \big| \sum m_I \underline{\dim} I = \beta\}$$

where $\mathrm{Indep}(\vec{Q})$ *denotes the set of indecomposables in* $\mathrm{Rep}^{nil}(\vec{Q})$ *and* $\underline{\dim} I$ *is the dimension vector of* I.

Proof. The first identity comes form the PBW theorem. The second holds up for $\mathrm{Rep}^{nil}(\vec{Q})$ is Krull-Schmit. □

With the claim above, by Gabriel 5.18, $\dim(\tilde{\mathbf{H}}_{\vec{Q}})_\beta = (\mathbf{U}_v(\mathfrak{b}_+))_\beta$, which implies Φ is a surjection. □

Remark. In this proof, we can see that $\Phi|_{\mathbf{U}_v(\mathfrak{n}_+)} : \mathbf{U}_v(\mathfrak{n}^+) \longrightarrow \tilde{\mathbf{H}}_{\vec{Q}}$ is also an isomorphism of Hopf algebra.

Example 6.9. For quivers of finite type, $\Phi : \mathbf{U}_v(\mathfrak{b}_+) \longrightarrow \tilde{\mathbf{H}}_{\vec{Q}}$ is an isomorphism. However, for K_2, it is only an injection.

First all, the Cartan matrix of K_2 is

$$\begin{pmatrix} 2 & -2 \\ -2 & 2 \end{pmatrix}$$

which is not positive definite. Hence by 5.18, K_2 is not a quiver of finite type.

The simple roots of $\mathfrak{g}_{K_2}$ (in fact, it is $\hat{\mathfrak{sl}}_2$) are α_1, α_2 and

$$\Delta^+ = \{n\alpha_1 + (n+1)\alpha_2, (n+1)\alpha_1 + n\alpha_2, n\alpha_1 + n\alpha_2 \big| n \geq 2\}$$

PBW theorem implies that

$$\dim(\mathbf{U}_v(\mathfrak{n}_+))_{(1,1)} = 2$$

However, we can directly compute representations with $\underline{\dim} V = (1,1)$. Consider

$$\bullet \underset{h_2}{\overset{h_1}{\rightrightarrows}} \bullet$$

If $h_1, h_2 \neq 0$, then we can multiply h_1 by $\lambda \in k^*$ such that $h_1 = \mathrm{id}$ and so $h_2 = \lambda \in k^*$. The number of such representations are $q-1$. Then we consider $h_1 = 0, h_2 \neq 2$, $h_1 \neq 0, h_2 = 0$ and $h_1 = 0, h_2 = 0$. Except the last case, all representations are indecomposable. For the last case, it is product of S_1 and S_2. Thus,

$$\dim(\tilde{\mathbf{H}}_{\vec{Q}})_{(1,1)} = q + 2$$

The graded structures of $\mathbf{U}_v(\mathfrak{b}_+)$ and $\tilde{\mathbf{H}}_{\vec{Q}}$ are different. Thus, it can not be an isomorphism.

7 PBW Theorem

7.1 Braiding Group

Let R be a finite root system with a set of simple roots $\Delta = \{\alpha_1, \alpha_2, \cdots, \alpha_n\}$. For each root α, define the reflection

$$s_\alpha(v) = v - \frac{2(\alpha, v)}{(\alpha, \alpha)}\alpha$$

The **Weyl group** W of R is the subgroup of orthogonal group generated by teh reflections corresponding to the simple roots, that is,

$$W = \langle s_{\alpha_1}, ..., s_{\alpha_n} \rangle$$

The **length** of $w \in W$ is

$$l(w) := \min\{k \big| w = s_{i_1} s_{i_2} \cdots s_{i_k}\}$$

the minimal number of simple reflections needed to write w.

Example 7.1. For A_{n-1}, the Weyl group is just S_n. Consider

$$\Delta = \{e_i - e_j \big| i \neq j\}$$

Then the simple roots of A_n are $\alpha_i = e_i - e_{i+1}$ with $i = 1, ..., n-1$. Then

$$s_{\alpha_i}(v) = v - (x_i - x_j)(e_i - e_j)$$

with $x_i - x_j = (v, \alpha_i)$, the inner product. Thus, for each v, s_{α_i} replaces x_i by x_j and replace x_j by x_i. Hence, s_{α_i} corresponding $(i, i+1)$, which implies $W = S_n$.

For simple reflections s_i, s_j corresponding to simple roots α_i, α_j, let m_{ij} be the order of $s_i s_j$ in the Weyl group W. Then

Definition 7.2. The **Braid group** B associated with the same root system is given by generators: $\sigma_1, ..., \sigma_n$ with relations

$$\underbrace{\sigma_i \sigma_j \sigma_i \cdots}_{m_{ij}} = \underbrace{\sigma_j \sigma_i \sigma_j \cdots}_{m_{ij}}$$

Example 7.3. For A_2, we have simple reflections s_1, s_2 with $s_1^2 = 1, s_2^2 = 1$ and $(s_1 s_2)^3 = 1$. (Draw the picture!) Thus,

$$B_{A_2} = \langle \sigma_1, \sigma_2 \big| \sigma_1 \sigma_2 \sigma_1 = \sigma_2 \sigma_1 \sigma_2 \rangle$$

For each $a \in \mathbb{C}$, $x \in \mathfrak{g}$, we formally write

$$\exp(a \cdot \mathrm{ad}) = \sum_{m=0}^{\infty} \frac{(a \cdot \mathrm{ad})^m}{m!}$$

Given a Lie algebra with generators e_i, f_i, h_i, define

$$t_i = \exp(\mathrm{ad}\ e_i) \circ \exp(-\mathrm{ad}\ f_i) \exp(\mathrm{ad}\ e_i)$$

automorphisms of $\mathfrak{g}$. Then we can extend them to automorphisms of $\mathbf{U}(\mathfrak{g})$.

Theorem 7.4. *Let $\mathfrak{g}$ and $\mathbf{U}(\mathfrak{g})$ be a Lie algebra and the corresponding enveloping algebra with Cartan matrix of $\mathfrak{g}$ of finite type. Then there is a group homomorphism*

$$\begin{aligned} B &\longrightarrow \mathrm{Aut}(\mathfrak{g}) \\ \sigma_i &\longmapsto t_i \end{aligned}$$

and hence a group homomorphism

$$\begin{aligned} B &\longrightarrow \mathrm{Aut}(U(\mathfrak{g})) \\ \sigma_i &\longmapsto t_i \end{aligned}$$

That is, the automorphisms t_i satisfying the braid relations.

Without proof, we will give an example to understand this.

Example 7.5. For A_2,

$$\begin{aligned} t_1(e_1) &= \exp(\mathrm{ad}\ e_1) \circ \exp(-\mathrm{ad}\ f_1) \circ \exp(\mathrm{ad}\ e_1)(e_1) \\ &= \exp(\mathrm{ad}\ e_1) \circ \exp(-\mathrm{ad}\ f_1)(e_1) \\ &= \exp(\mathrm{ad}\ e_1)(e_1 + \sum_{k=1}^{\infty}(-1)^k \frac{\mathrm{ad}^k\ f_1(e_1)}{k!}) \\ &= \exp(\mathrm{ad}\ e_1)(e_1 + h_1 - f_1) \\ &= -f_1 \end{aligned}$$

Now, let's consider the quantum group $\mathbf{U}_v(\mathfrak{g})$. For each $i \in \{1, ..., n\}$, there is a unique $\mathbb{Q}(q)$ algebra isomorphism

$$T_i : \mathbf{U}_q(\mathfrak{g}) \longrightarrow \mathbf{U}_q(\mathfrak{g})$$

such that

$$T_i(E_j) = \begin{cases} -F_jK_j, \text{ if } i = j; \\ E_j, \text{ if } a_{ij} = 0; \\ -E_iE_j + q^{-1}E_jE_i, \text{ if } a_{ij} = -1. \end{cases} \quad ; \quad T_i^{-1}(E_j) = \begin{cases} -K_j^{-1}F_j, \text{ if } i = j; \\ E_j, \text{ if } a_{ij} = 0; \\ -E_jE_i + q^{-1}E_iE_j, \text{ if } a_{ij} = -1. \end{cases} .$$

$$T_i(F_j) = \begin{cases} -K_j^{-1}E_j, \text{ if } i = j; \\ F_j, \text{ if } a_{ij} = 0; \\ -F_jF_i + qF_iF_j, \text{ if } a_{ij} = -1. \end{cases} \quad ; \quad T_i^{-1}(F_j) = \begin{cases} -E_jK_j, \text{ if } i = j; \\ F_j, \text{ if } a_{ij} = 0; \\ -F_iF_j + qF_jF_i, \text{ if } a_{ij} = -1. \end{cases} ;$$

$$T_i(K_j) = \begin{cases} K_j^{-1}, \text{ if } i = j; \\ K_j, \text{ if } a_{ij} = 0; \\ K_iK_j, \text{ if } a_{ij} = -1. \end{cases} \quad ; \quad T_i(K_j) = \begin{cases} K_j^{-1}, \text{ if } i = j; \\ K_j, \text{ if } a_{ij} = 0; \\ K_iK_j, \text{ if } a_{ij} = -1. \end{cases}$$

Remark. Here, we only consider the Kac-Moody algebra with Cartan algebra $A = (a_{ij})$ such that $-1 \leq a_{ij} \leq 0$ when $i \neq j$.

Note that $T_iT_jT_i = T_jT_iT_j$ if $a_{ij} = -1$ and $T_iT_j = T_jT_i$ if a_{ij}. Thus, defines a homomorphism of B of W, the Weyl group of $\mathfrak{g}$ into $\mathrm{Aut}(\mathbf{U}_q(\mathfrak{g}))$.

Remark. For any Cartan matrix $A = (a_{ij})$, the order m_{ij} of s_is_j in depends only on the product of a_{ij}, a_{ji}. For example,

$$m_{ij} = \begin{cases} 2, \text{ if } a_{ij}a_{ji} = 0; 3, \text{ if } a_{ij}a_{ji} = 1; 4, \text{ if } a_{ij}a_{ji} = 2; 6, \text{ if } a_{ij}a_{ji} = 3. \end{cases}$$

Thus, we just need $T_iT_jT_i = T_jT_iT_j$ if $a_{ij} = -1$ and $T_iT_j = T_jT_i$ if a_{ij} to give the homomorphism from B to $W_{\mathfrak{g}}$.

7.2 PBW Theorem

For any $w = s_{i_1} s_{i_2} \cdots s_{i_n}$, define

$$T_w = T_{i_1} \cdots T_{i_n}$$

Then

Proposition 7.6. *Let $w \in W$. Then*

$$T_w(\mathbf{U}_q^{res}(\mathfrak{g})) \subseteq \mathbf{U}_q^{res}(\mathfrak{g})$$

Proof. It is enough to show that $T_i(\mathbf{U}) \subseteq \mathbf{U}$ and $T_i^{-1}(\mathbf{U}) \subset \mathbf{U}$ for any $i \in \{1, ..., n\}$. □

Now, we construct a putative basis of $\mathbf{U}_q(\mathfrak{g})$, using braiding group action: We choose for each $\beta \in R^+$ an element $\omega_\beta \in W$ for some $i_\beta \in \{1, ..., n\}$,

$$\omega_\beta^{-1}(\beta) = \alpha_{i_\beta}$$

Let $\mathbb{N}^{R^+}$ be the set of all functions $R^+ \longrightarrow \mathbb{N}$, we fix a total order on R^+ and define for each $\Psi \in \mathbb{N}^{R^+}$:

$$E^\Psi = \Pi_{\beta \in R^+} T_{\omega_\beta}(E_{i\beta})^{\Psi(\beta)}, F^\Psi = \Pi_{\beta \in R^+} T_{\omega_\beta}(F_{i\beta})^{\Psi(\beta)}$$

Proposition 7.7. *E^Ψ (resp. F^Ψ) for $\Psi \in \mathbb{N}^{R^+}$ are linearly independent in the $\mathbb{Q}(q)$-vector space $\mathbf{U}_q^+(\mathfrak{g})$.*

Let $\mathbf{U}_q^0$ be the subalgebra of $\mathbf{U}_q(\mathfrak{g})$ generated by K_i and K_i^{-1} with $1 \leq i \leq n$. For $\Phi = \sum \Phi(i)\alpha_i \in \mathbb{Z}^n$, define

$$K^\Phi = K_1^{\Phi(1)} K_2^{\Phi(2)} \cdots K_n^{\Phi(n)} \in \mathbf{U}_q^0$$

Lemma 7.8. *(a). Multiplication defines*

$$\mathbf{U}_q^-(\mathfrak{g}) \otimes_{\mathbb{Q}(q)} \mathbf{U}_q^0(\mathfrak{g}) \otimes_{\mathbb{Q}(q)} \mathbf{U}^+(\mathfrak{g}) \xrightarrow{\cong} \mathbf{U}_q(\mathfrak{g})$$

(b). $K^\Phi, \Phi \in \mathbb{Z}$ form a $\mathbb{Q}(q)$-basis of $\mathbf{U}_q^0(\mathfrak{g})$.

Theorem 7.9 (PBW)**.** *The elements $F^\Psi K^\Phi E^{\Psi'}$ with $\Psi', \Psi \in \mathbb{N}^{R^+}, \Phi \in \mathbb{Z}^n$ form a $\mathbb{Q}(q)$-basis of $\mathbf{U}_q(\mathfrak{g})$.*

8 Drinfield Double

8.1 Hopf Algebra

Let's first recall some basic concepts of algebras and coalgebras.

Definition 8.1 (Algebra)**.** An algebra is given by a triple (A, μ, η) with A a vector space, $\mu : A \otimes A \longrightarrow A$ and $\eta : k \longrightarrow A$ linear maps satisfying

$$(\text{Ass}) : \mu \circ (\mu \otimes \text{id}) = \mu \circ (\text{id} \otimes \mu)$$

$$(\text{Unit}) : \text{can} = \mu \circ (\eta \otimes \text{id}),\ \text{can} = \mu \circ (\text{id} \otimes \eta)$$

where $\text{can} : k \otimes A \xrightarrow{\cong} A$. Moreover, we say (A, μ, η) is commutative if

$$(\text{Comm}) : \mu = \mu \circ \tau_{A,A}$$

A linear map $f : (A, \mu, \eta) \longrightarrow (A', \mu', \eta')$ is a morphism of algebras if

$$\mu' \circ (f \otimes f) = f \circ \mu,\ f \circ \eta = \eta'$$

Definition 8.2 (Coalgebra)**.** A coalgebra is a triple (C, Δ, ε), with C a vector space, $\Delta : C \longrightarrow C \otimes C$ and $\varepsilon : C \longrightarrow k$ linear maps satisfying

$$(\text{Coass}) : (\Delta \otimes \text{id}) \circ \Delta = (\text{id} \otimes \Delta) \circ \Delta$$

$$(\text{Counit}) : \text{can} \cong (\varepsilon \otimes \text{id}) \circ \Delta,\ \text{can} \cong (\text{id} \otimes \varepsilon) \circ \Delta$$

Moreover, (C, Δ, ε) is cocommutative if

$$(\text{Cocomm}) : \Delta = \Delta \circ \tau_{C,C}$$

A linear map $f : (C, \Delta, \varepsilon) \longrightarrow (C', \Delta', \varepsilon')$ is called a morphism of coalgebras if

$$(f \otimes f) \circ \Delta = \Delta' \circ f,\ \varepsilon = \varepsilon' \circ f$$

Example 8.3 (Algebra and coalgebra structure over $k[x]$. $]For\text{k[x]}, itshasthestructureofanalgebragivenby\mu$: $f(x) \otimes g(x) \mapsto f(x)g(x);\ \eta : 1 \mapsto 1$ It is easy to verify that $k[x]$ satisfies the (Ass), (Unit) above. It is also a coalgebra via:

$$\Delta(f(x)) \mapsto f(x \otimes 1 + 1 \otimes x);\ \varepsilon : f(x) \mapsto f(0)$$

where the multiplication over $k[x] \otimes k[x]$ is induced by μ. For example,

$$\Delta(1) = 1 \otimes 1,\ \Delta(x) = x \otimes 1 + 1 \otimes x,\ \Delta(x^2) = x^2 \otimes 1 + 2x \otimes x + 1 \otimes x^2.$$

Since Δ, $\Delta \otimes \text{id}$ are in fact morphisms between rings and linear, to check that it satisfies (Coass), (Counit), it is enough to check everything for x, which is obviously.

Now, for $k[x, y]$, it is obviously an algebra. It is also a coalgebra given by

$$\Delta : f(x, y) \mapsto f(x \otimes 1 + 1 \otimes x, y \otimes 1 + 1 \otimes y);\ \eta : f(x, y) \mapsto f(0, 0).$$

For example,

$$\Delta(x, y) = xy \otimes 1 + x \otimes y + y \otimes x + 1 \otimes xy$$

Lemma 8.4. *The tensor product $C \otimes C'$ of two coalgebras (C, Δ, ε) and $(C', \Delta', \varepsilon')$ has a coalgebra structure with comultiplication $(\text{id} \otimes \tau_{C,C'} \otimes \text{id}) \circ (\Delta \otimes \Delta')$ and counit $\varepsilon \otimes \varepsilon'$.*

Proof. Take $c, c' \in C, C'$. Suppose that $\Delta(c) = \sum c_{(1)} \otimes c_{(2)}$ and $\Delta(c') = \sum c'_{(1)} \otimes c'_{(2)}$. Then

$$\Delta(c \otimes c') = \sum c_{(1)} \otimes c'_{(1)} \otimes c_{(2)} \otimes c'_{(2)}$$

Now, we check the (Coass):

$$(\Delta \otimes \mathrm{id}) \circ \Delta(c \otimes c') = \sum c_{(11)} \otimes c'_{(11)} \otimes c_{(12)} \otimes c'_{(12)} \otimes c_{(2)} \otimes c'_{(2)}$$

$$(\mathrm{id} \otimes \Delta) \circ \Delta(c \otimes c') = \sum c_{(1)} \otimes c'_{(1)} \otimes c_{(21)} \otimes c'_{(21)} \otimes c_{(22)} \otimes c'_{(22)}$$

Since C and C' are coalgebras, we have

$$\sum c_{(1)} \otimes c_{(21)} \otimes c_{(22)} = \sum c_{(1)} \otimes c_{(21)} \otimes c_{(22)}$$

$$\sum c'_{(1)} \otimes c'_{(21)} \otimes c'_{(22)} = \sum c'_{(1)} \otimes c'_{(21)} \otimes c'_{(22)}$$

which implies $(\Delta \otimes \mathrm{id}) \circ \Delta = (\mathrm{id} \otimes \Delta) \circ \Delta$.

For the (counit), it also comes form the counity of C and C'. We skip the proof. □

Here is an example to see why the comultiplication and counit are defined as in 8.4:

Example 8.5. Recall in 8.3, $k[x, y] = k[x] \otimes k[y]$ and $k[x], k[y]$ are both coalgebras. With the comultiplication given in 8.4, we have

$$\begin{aligned}\Delta(x \otimes y) &= \mathrm{id} \otimes \tau_{x,y} \otimes \mathrm{id}(1 \otimes x \otimes 1 \otimes y + x \otimes 1 \otimes 1 \otimes y + 1 \otimes x \otimes y \otimes 1 + x \otimes 1 \otimes y \otimes 1) \\ &= 1 \otimes 1 \otimes x \otimes y + x \otimes 1 \otimes 1 \otimes y + 1 \otimes y \otimes x \otimes 1 + x \otimes y \otimes 1 \otimes 1\end{aligned}$$

We can see under the morphism:

$$k[x] \otimes k[y] \longrightarrow k[x, y], f(x) \otimes g(y) \mapsto f(x)g(y)$$

$k[x] \otimes k[y]$ and $k[x, y]$ are just isomorphic.

Let H be a vector space equipped with an algebra structure (H, μ, η) and a coalgebra structure (H, Δ, ε). Then we can assign $H \otimes H$ the induced structure of algebras and coalgebras by 8.4.

Theorem 8.6. *The followings are equivalent:*

(i). The map μ and η are morphisms of algebras;

(ii). Δ and ε are morphisms of coalgebras.

Proof. to be continued □

Definition 8.7 (Bialgebra)**.** A bialgebra is a quintuple $(H, \mu, \eta, \Delta, \varepsilon)$ where (H, μ, η) is an algebra and (H, Δ, ε) is a coalgebra, verifying the equivalent conditions of the theorem above.

A morphism of bialgebras is a morphism for the underlying algebra and coalgebra structures.

Given an algebra (A, μ, η) and a coalgebra (C, Δ, ε), we can define a bilinear map, the convolution, on the vector space $\mathrm{Hom}(C, A)$. By defintion, for $f, g \in \mathrm{Hom}(C, A)$, $f * g$ is given by

$$C \xrightarrow{\Delta} C \otimes C \xrightarrow{f \otimes g} A \otimes A \xrightarrow{\mu} A$$

Definition 8.8 (Hopf algebra)**.** Let $(H, \mu, \eta, \Delta, \varepsilon)$ be a bialgebra. An endomorphism S of H is called an antipode for the bialgebra H if

$$S * \mathrm{id} = \mathrm{id} * S = \eta \circ \varepsilon$$

A Hopf algebra is a bialgebra with an antipode and a morphism of Hopf algebras is a morphism between the underlying bialgebras commuting with the antipodes.

Given a Hopf algebra $(H, \mu, \eta, \Delta, \varepsilon, S, S^{-1})$, there is another Hopf algebra given by it in a canonical way:

Theorem 8.9. *If $(H, \mu, \eta, \Delta, \varepsilon, S, S^{-1})$ is a finite-dimension Hopf algebra. Then*

$$X = (H^{op})^* := (H^*, \Delta^*, \varepsilon, (\mu^{op})^*, \eta, (S^{-1})^*, S^*)$$

is a Hopf algerba.

Proof. To be continued □

8.2 Bicrossed Products

Definition 8.10. Let H be a bialgebra and C be a colagebra. We say that C os a mod-coalgebra over H if there exists a morphism of coalgebras:

$$H \otimes C \longrightarrow C$$

inducing an H-mod structure on C.

Example 8.11. Let G be a finite group and $H = kG$ be a group algebra. Then H is a coalgebra with

$$\Delta(g) = g \otimes g, \varepsilon : g \mapsto 1$$

Moreover, choose X to be a finite set and set $C = kX$. Over C, then there exists a natural coalgebra structure over $C^* = \mathrm{Hom}(C, k)$:

$$\Delta(f)(x, y) = \begin{cases} f(x), x = y; \\ 0, \quad x \neq y. \end{cases}$$

Suppose that G acts on X. Define

$$(g \cdot f)(x) = f(g^{-1}x)$$

Then C^* is a kG-mod coalgebra: Take $g \in H$ and $\mathbf{1}_x \in C^*$

$$\Delta(g \otimes \mathbf{1}_x) = g \otimes \mathbf{1}_x \otimes g \otimes \mathbf{1}_x$$

and under the map, $\Delta(g \otimes \mathbf{1}_x)$ is mapped to $\mathbf{1}_{g_x} \otimes \mathbf{1}_{gx}$ over $C^* \otimes C^*$, which is just $\Delta(g \cdot \mathbf{1}_x)$.

Definition 8.12. A pair (X, A) of bialgebra is mathched if there exist linear maps $\alpha : A \otimes X \longrightarrow X$ and $\beta : A \otimes X \longrightarrow A$, turning X into a mod-coalgebra over A, and turning A into a mod-coalgebra over X, such that, if we set

$$\alpha(a \otimes x) = a \cdot x; \ \beta(a \otimes x) = a^x$$

the followings are satisfied:

$$a \cdot (xy) = \sum (a_{(1)} \cdot x_{(1)})(a_{(2)}^{x_{(2)}} y), \ a \cdot 1 = \varepsilon(a)1;$$

$$(ab)^x = \sum a^{b_{(1)} \cdot x_{(1)}} b_{(2)}^{x_{(2)}}, \ 1^x = \varepsilon(x)1$$

$$\sum a_{(1)}^{x_{(1)}} \otimes (a_{(2)} \cdot x_{(2)}) = \sum a_{(2)}^{x_{(2)}} \otimes (a_{(1)} \cdot x_{(1)})$$

Theorem 8.13. *Let (X, A) be a matched pair of bialgebras. There exists a unique bialgebra structure on the vector space $X \otimes A$, with unit $1 \otimes 1$, counit: $\varepsilon(x \otimes a) = \varepsilon(x) \cdot \varepsilon(a)$, product:*

$$(x \otimes a) \cdot (y \otimes b) = \sum x(a_{(1)} \cdot y_{(1)}) \otimes a_{(2)}^{y_{(2)}} b$$

and coproduct

$$\Delta(x \otimes a) = \sum (x_{(1)} \otimes a_{(1)}) \otimes (x_{(2)} \otimes a_{(2)})$$

for any $x, y \in X$ and $a, b \in A$. Equipped with this bialgebra strutcre, $X \otimes A$ is called the bicrossed product of X and A, and denoted by $X \bowtie A$.

Furthermore, if X, A have antipodes, denoted by S_X, S_A, then

$$S(x \otimes a) = \sum S_A(a_{(2)}) \cdot S_X(x_{(2)}) \otimes S_A(a_{(1)})^{S_X(x_{(1)})}$$

is an antipodes over $X \bowtie A$.

Proof. **Setp 1**: Verify (Ass), (Un). Note that Coass, Coun are just the same as the coalgebra structure over $X \otimes A$, which is verified in 8.4.

First, we compute $((x \otimes a) \cdot (y \otimes b)) \cdot (z \otimes c)$:

$$\begin{aligned}
((x \otimes a) \cdot (y \otimes b)) \cdot (z \otimes c) &= \sum (x(a_{(1)} \cdot y_{(1)}) \otimes (a_{(2)}^{y_{(2)}} \cdot b) \cdot (z \otimes c) \\
&= \sum (x(a_{(1)} \cdot y_{(1)})(a_{(21)}^{y_{(21)}} b_{(1)} \cdot z_{(1)})) \otimes ((a_{(22)}^{y_{(22)}} b_{(2)})^{z_{(2)}} c) \\
&= \sum (x(a_{(1)} \cdot y_{(1)})(a_{(21)}^{y_{(21)}} b_{(1)} \cdot z_{(1)})) \otimes ((a_{(221)}^{y_{(221)}})^{b_{(21)} \cdot z_{(21)}}) b_{(22)}^{z_{(22)}} c) \\
&= \sum x(a_{(1)} \cdot y_{(1)})(a_{(2)}^{y_{(2)}} b_{(1)} \cdot z_{(1)}) \otimes (((a_{(3)})^{y_{(3)}})^{b_{(2)} \cdot z_{(2)}} b_{(3)}^{z_{(3)}}) c)
\end{aligned}$$

where the third equality holds up since (X, A) is matched, and the last equality holds up since $\Delta^2 = (\mathrm{id} \otimes \Delta)\Delta$ is welled-defined. For $(x \otimes a) \cdot ((y \otimes b) \cdot (z \otimes c))$, we have

$$\begin{aligned}
(x \otimes a) \cdot ((y \otimes b) \cdot (z \otimes c)) &= (x \otimes a)(\sum y(b_{(1)} \cdot z_{(1)}) \otimes (b_{(2)}^{z_{(2)}} c)) \\
&= \sum x(a_{(1)} \cdot (y_{(1)}(b_{(11)} \cdot z_{(11)}))) \otimes a_{(2)}^{y_{(2)}(b_{(12)} \cdot z_{(12)})} b_{(2)}^{z_{(2)}} c \\
&= \sum x(a_{(11)} \cdot y_{(11)})(a_{(12)}^{y_{(12)}} \cdot (b_{(11)} \cdot z_{(11)})) \otimes a_{(2)}^{y_{(2)}(b_{(12)} \cdot z_{(12)})} b_{(2)}^{z_{(2)}} c \\
&= \sum x(a_{(1)} \cdot y_{(1)})(a_{(2)}^{y_{(2)}} \cdot (b_{(1)} \cdot z_{(1)})) \otimes a_{(2)}^{y_{(3)}(b_{(2)} \cdot z_{(2)})} b_{(3)}^{z_3} c
\end{aligned}$$

where the third equality holds up since (X, A) is matched, and the last equality holds up since $\Delta^2 = (\Delta \otimes \mathrm{id})\Delta$. Thus, we can see that $\cdot$ defined over $X \otimes H$ satisfies (Ass).

For (Unit),

$$(1 \otimes 1) \cdot (x \otimes a) = \sum 1 \cdot x_{(1)} \otimes 1^{x_{(2)}} \cdot a = \sum x_{(1)} \otimes \varepsilon(x_{(2)}) a = x \otimes a$$

$$(x \otimes a) \cdot (1 \otimes 1) = \sum x \cdot (a_{(1)} \cdot 1) \otimes a_{(2)}^{1} \cdot 1 = \sum \varepsilon(a_{(1)}) x \otimes a_{(2)} = x \otimes a$$

Step 2: Show that coproduct and counit are morphisms of algebras, so that $X \otimes H$ is a bialgebra equipped morphisms above.

Step 3: Check that S is an antipode.

$$\begin{aligned}
\sum (x_{(1)} \otimes a_{(1)}) \cdot S(x_{(2)} \otimes a_{(2)}) &= \sum (x_{(1)} \otimes a_{(1)}) \cdot (S_A(a_{(22)}) \cdot S_X(x_{(22)}) \otimes S_A(a_{(21)})^{S_X(x_{21})}) \\
&= \sum x_{(1)}(a_{(11)}(S_A(a_{(222)}) \cdot S_X(x_{222}))) \otimes a_{(12)}^{S_A(a_{221}) \cdot S_X(x_{221})} S_A(a_{(21)})^{S_X(x_{(21)})} \\
&= \sum x_{(1)}(a_{(11)}(S_A(a_{(22)}) \cdot S_X(x_{22}))) \otimes (a_{(12)} S_A(a_{(21)})^{S_X(x_{(21)})} \\
&= \sum x_{(1)}(a_{(11)}(S_A(a_{(22)}) \cdot S_X(x_{22}))) \otimes \varepsilon(a_{(12)}) \varepsilon(S_X(x_{(21)})) \cdot 1
\end{aligned}$$

$$= \sum x_{(1)}(a_{(1)}(S_A(a_{(3)}) \cdot \varepsilon(S_X(x_{(2)})))) \otimes \varepsilon(a_{(2)}) \cdot 1$$
$$= \sum \varepsilon(x) \cdot \varepsilon(a) 1 \otimes 1$$

where we use the following diagrams of Δ^3:

$$(\mathrm{id} \otimes \mathrm{id} \otimes \Delta) \circ (\mathrm{id} \otimes \Delta) \circ \Delta : \begin{array}{cccc} a_{(1)} & a_{(2)} & & \\ a_{(1)} & a_{(21)} & a_{(22)} & \\ a_{(1)} & a_{(21)} & a_{(221)} & a_{(222)} \end{array}$$

and

$$(\mathrm{id} \otimes \mathrm{id} \otimes \Delta) \circ (\Delta \otimes \mathrm{id}) \circ \Delta : \begin{array}{cccc} a_{(1)} & a_{(2)} & & \\ a_{(11)} & a_{(12)} & a_{(2)} & \\ a_{(11)} & a_{(12)} & a_{(21)} & a_{(22)} \end{array}$$

You need to be careful for the third equality, where we use the first diagram and the matched conditions to reduce all things to the second row of the diagram. In the fifth equation, use $\Delta^2 a = \sum a_{(1)} \otimes a_{(2)} \otimes a_{(3)}$. Similarly, we can show that $\sum S(x_{(1)} \otimes a_{(1)}) \cdot x_{(2)} \otimes a_{(2)} = \varepsilon(x \otimes a) 1 \otimes 1$. Thus, S defined above is an antipode. □

Theorem 8.14. *Consider $(H, \mu, \eta, \Delta, \varepsilon, S, S^{-1})$ and $X = (H^{op})^*$. Let $\alpha : H \otimes X \longrightarrow X$ and $\beta : H \otimes X \longrightarrow X$ be linear maps given by*

$$\alpha(a \otimes f)(b) = \sum f(S^{-1}(a_{(1)}) b a_{(2)}), \ \beta(a \otimes f) = \sum f(S^{-1}(a_{(3)}) a_{(1)}) a_{(2)}$$

where $\Delta^2 a = \sum a_{(1)} \otimes a_{(2)} \otimes a_{(3)}$. Then the pair (H, X) is mathed. Moreover, $X \bowtie H$ is a Hopf algebra, called Drinfield double, denoted by $D(H)$.

Proof. To be continued □

8.3 Representation of Drinfield Double

We assign to any strict tensor category $(\mathcal{C}, \otimes, I)$ with a braided tensor categroy $\mathcal{Z}(\mathcal{C})$.

Definition 8.15. An object of $\mathcal{Z}(\mathcal{C})$ is a pair $(V, c_{_\otimes V})$, where V is an object of $\mathcal{C}$ and $c_{_,V}$ is a family of natural isomorphisms

$$c_{X,V} : X \otimes V \longrightarrow V \otimes X$$

defined for all objects X in $\mathcal{C}$ such that for any objects X, Y in $\mathcal{C}$, we have

$$c_{X \otimes Y, V} = (c_{X,V} \otimes \mathrm{id}_Y) \circ (\mathrm{id}_X \otimes c_{Y,V})$$

A morphism from $(V, c_{_,V})$ is a morphism $f : V \longrightarrow W$ in $\mathcal{C}$ such that for each object X of $\mathcal{C}$, we have

$$(f \otimes \mathrm{id}_X) c_{X,V} = c_{X,W} (\mathrm{id}_X \otimes f)$$

Example 8.16. To be continued

Theorem 8.17. *Let $(\mathcal{C}, \otimes, I)$ be a strict tensor categroy. Then $\mathcal{Z}(\mathcal{C})$ is a strict braided tensor category, where*

(i). the unit is (I, id);

(ii). the tensor product of $(V, c_{_,V})$ and $(W, c_{_,W})$ is given by

$$(V, c_{_,V}) \otimes (W, c_{_,W}) = (V \otimes W, c_{_,V \otimes W})$$

where $c_{X,V \otimes W} : X \otimes V \otimes W \longrightarrow V \otimes W \otimes X$ is the morphism of $\mathcal{C}$ defined for any objects X in $\mathcal{C}$ by

$$c_{X,V \otimes W} = (\mathrm{id}_V \otimes c_{X,W}) \circ (c_{X,V} \otimes \mathrm{id}_W)$$

(iii). The braiding is given by

$$c_{V,W} : (V, c_{_,V}) \otimes (W, c_{_,W}) \longrightarrow (W, c_{_,W}) \otimes (V, c_{_,V})$$

Theorem 8.18. *For any finite-dimensional Hopf algebra H, with invertible antipode, the braided tensor categories $\mathcal{Z}(A\text{-}Mod)$ and $\mathcal{D}(A)$-Mod are equivalent.*

Using Drinfield double, we can give a description of $\mathrm{U}_v(\mathfrak{g})$ via $U_v(\mathfrak{b})$ for any Kac-Moody algebra $\mathfrak{g}$:

Proposition 8.19. *Let $K = \mathbb{C}(v)[K_i^{\pm}]_{i\in I} \subseteq \mathbf{U}_v(\mathfrak{b}_+)$ be the quantized enveloping algebra of the Cartan subalgebra $\mathfrak{h} \subset \mathfrak{b}_+$. Then $K_i \otimes K_i^{-1} \in D\mathbf{U}_v(\mathfrak{b}_+)$ are the central elements for all i, and there is an isomorphisms of Hopf algebra:*

$$D\mathbf{U}_v(\mathfrak{b}_+)/\langle K_i \otimes K_i^{-1} - 1\rangle_i \cong \mathbf{U}_v(\mathfrak{g})$$

9 The representation of quivers over $\mathbb{F}_1$

Over a finite field $\mathbb{F}_q$, we already know some properties about the linear algebras over $\mathbb{F}_q$. But, what's will happen when taking the limit $q \longrightarrow 1$? First let's consider the following example:

Example 9.1. In the computation before, we have seen that

$$\#\mathrm{Gr}(k,n) = \frac{[n]_q!}{[k]_q![n-k]_q!}$$

while over $\mathbb{F}_q$, $[n]_q = 1 + q + \cdots + q^{n-1}$. It counts the number of subspaces of $\mathbb{F}_q^n$ of dimension k. If taking the limit $q \longrightarrow 1$, this reduces to the binormal coefficient $\binom{n}{k}$, counting k-elements subsets of an n-element set.

9.1 The category Vect($\mathbb{F}_1$)

Definition 9.2. The category $\mathrm{Vect}(\mathbb{F}_1)$ of vector spaces over $\mathbb{F}_1$ is defined with objects pointed finite sets $(V, \mathcal{O}_V)$ and $\mathrm{Hom}(V,W)$ the set of maps $f : V \longrightarrow W$ such that $f(0_V) = 0_W$ and $f|_{V \setminus f^{-1}(0_W)}$ is an injection.

Remark. i). Composition of morphisms is defined as the composition of maps of sets and the zero map is just $f : V \longrightarrow W$, $f(V) = 0_W$.

ii). We can define $\dim V = V - 1$, which implies that we can think that the pointed element in the set as the zero vector and the other elements as basis vectors.

$\mathrm{Vect}(\mathbb{F}_1)$ is an abelian category: For $X \subset Y$, denote by Y/X the pointed set obtained by identifying X to a single point. Then we can construct direct sum, ker, coker, etc. over $\mathrm{Vect}(\mathbb{F}_1)$ as follows:

i). The unique object of $\mathrm{Vect}(\mathbb{F}_1)$ of dimension 0 is an initial, terminal, and hence zero object;

ii). For any $f \in \mathrm{Hom}(V,W)$, $\ker f = f^{-1}(0_W)$ and $\mathrm{coker}\ f = W/f(V)$;

iii). $V \oplus W = V \sqcup U/\{0_V, 0_W\}$ and there is $l_V : V \hookrightarrow V \oplus W$ and $p_V : V \oplus W \twoheadrightarrow V$ such that $p_V \circ l_V = \mathrm{id}_V$;

iv). $\mathrm{Vect}(\mathbb{F}_1)$ has a symmetry monoidal structure on $V \otimes W$, defined as

$$V \otimes W := V \times W/\{V \times 0_W \cup 0_V \times W\}$$

and $(V \oplus W) \otimes Z \cong (V \otimes Z) \oplus (W \otimes Z)$, $Z \otimes (V \oplus W) \cong (Z \otimes V) \oplus (Z \otimes W)$;

v). A subsapce of V is a subset $U \subset V$, containing 0_V.

Here, we need some comments for $\mathrm{Ext}^n(V,W)$: $\mathrm{Ext}^n(V,W)$ is the set of equivalence classes of $(n+2)$-step exact sequences, that is,

$$\zeta : 0 \longrightarrow W \longrightarrow E_n \longrightarrow \cdots \longrightarrow E_1 \longrightarrow V \longrightarrow 0$$

ζ and ζ' are equivalent if there exists $\phi_k : E_k \xrightarrow{\cong} E'_k$ such that

$$\begin{tikzcd}
0 \arrow[r] & W \arrow[r] \arrow[d, equal] & E_n \arrow[r] \arrow[d, "\phi_n"] & \cdots \arrow[r] & E_1 \arrow[r] \arrow[d, "\phi_1"] & V \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & W \arrow[r] & E_n \arrow[r] & \cdots \arrow[r] & E_1 \arrow[r] & V \arrow[r] & 0
\end{tikzcd}$$

commutes.

Just as linear algebras over $\mathbb{F}_q$, we also care properties of $End(V)$ for $V \in \mathrm{Vect}(\mathbb{F}_1)$. It turns out that Jordan decomposition still holds up over $\mathrm{Vect}(\mathbb{F}_1)$:

Definition 9.3. Suppose that $V \in \mathrm{Vect}(\mathbb{F}_1)$ and $T \in End(V)$. We can write $T = \oplus_{j\in J} T_j$ if there exists a decomposition $V = \oplus_j V_j$ such that $T(V_j) \subset V_j$ and $T_j := T|_{V_j}$. In this case, we say that T is a direct sum of T_j.

We say T is equivalent to $T' \in End(W)$, if there exists an isomorphism $\sigma : V \longrightarrow W$ such that $T = \sigma^{-1}T'\sigma$.

Let $\mathbf{m} \in \mathrm{Vect}(\mathbb{F}_1)$ be $\{0_m, 1, 2, \cdots, m\}$. Let $N_m \in End(\mathbf{m})$ be

$$N_m(k) = k - 1,\ k = 1, \cdots, m.$$

and $C_m \in End(\mathbf{m})$ be

$$C_m(k) = k - 1,\ k = 2, \cdots, m;\ C_m(1) = m.$$

Then N_m is the analogue of a $m \times m$ nilpotent Jordan block, whereas C_m is a cyclic permutation of $\mathbf{m} \setminus \{0_m\}$. Just as what we have in the general linear algebra:

Lemma 9.4. *Every $T \in End(V)$ can be written as a direct sum $T = \oplus_j T_j$, where $T_j \in End(V_j)$ is equivalent to either N_{m_i} or C_{m_i} with $m_i = \dim V_i$. This decomposition is unique up to the ordering of factors.*

Proof. To be continued □

9.2 Representation over $\mathbb{F}_1$

Let $Q = (I, H)$ be a quiver.

Definition 9.5. A representation of a quiver Q in $\mathrm{Vect}(\mathbb{F}_1)$ is
i). An assignment of object $V_i \in \mathrm{Vect}(\mathbb{F}_1)$ to each $i \in I$;
ii). For each $h \in H$, a choice $f_h \in \mathrm{Hom}(V_{s(h)}, V_{t(h)})$.

Definition 9.6. The category $\mathrm{Rep}(Q, \mathbb{F}_1)$ is defined as follows:
i). Objects are representations V of Q in $\mathrm{Vect}(\mathbb{F}_1)$;
ii). $\mathrm{Hom}(V, W) := \{\Phi \in (\phi_i)_{i\in I}, \phi_i \in \mathrm{Hom}(V_i, W_i), g_h \circ \phi_{s(h)} = \phi_{t(h)} \circ f_h\}$.

In $\mathrm{Rep}(Q, \mathbb{F}_1)$, we can also get well-defined kernel, cokernel, direct sum, and so on. Also, we can define the category $\mathrm{Rep}^{nil}(Q, \mathbb{F}_1)$ and $\{S_i \big| i \in I\}$ are the only simple objects in $\mathrm{Rep}^{nil}(Q, \mathbb{F}_1)$. Moreover, we have the Jordan-Hölder theorem holding in $\mathrm{Rep}(Q, \mathbb{F}_1)$:

Theorem 9.7 (Jordan-Hölder)**.** *Every $V \in \mathrm{Rep}(Q, \mathbb{F}_1)$ processes a filtration*

$$0 = V_0 \subset V_1 \subset \cdots \subset V_n = V$$

such that $A_i = V_i/V_{i-1}$ is simple and if $0 = V_0' \subset V_1' \subset \cdots \subset V_n' = V$ is another such filtration with $B_i = V_i'/V_{i-1}'$, there exists a permutation σ of $1, \cdots, n$ such that $A_j = B_{\sigma(j)}$.

Proof. To be continued □

In fact, $\mathrm{Rep}(Q, \mathbb{F}_1)$ is Krull-Schmit.

Theorem 9.8 (Krull-Schmit)**.** *Every $V \in \mathrm{Rep}(Q, \mathbb{F}_1)$ can be uniquely (up to recording) decomposed into a direct sum of indecomposable representations.*

Proof. To be continued □

Since Jordan-Hölder holds in $\mathrm{Rep}(Q, \mathbb{F}_1)$ and $\{S_i \big| i \in I\}$ are the all simple objects in $\mathrm{Rep}(Q, \mathbb{F}_1)$, we can give the structure of Grothendieck group:

Corollary 9.9. *$K_0(\mathrm{Rep}(Q,\mathbb{F}_1)) \cong \mathbb{Z}^{|I|}$, spanned by the classes of the S_i, $i \in I$.*

Example 9.10. Consider the E_4 quiver.

$$\begin{tikzcd} 1 \arrow[r] & 2 & 3 \arrow[l] \\ & 4 \arrow[u] & \end{tikzcd}$$

We know that over $k = \mathbb{F}_q$, there exists an indecomposable:

$$E_{12^234}: \quad \begin{tikzcd} k \arrow[r, "(1,0)"] & k^2 & k \arrow[l, "(0,1)"'] \\ & k \arrow[u, "(1,1)"'] & \end{tikzcd}$$

However, over F_1, E_{12^234} is decomposable:

$$\begin{tikzcd} \{0,e_1\} \arrow[r] & \{0,e_1,e_2\} & \{0,e_2\} \arrow[l] \\ & \{0,v\} \arrow[u, "v\mapsto e_1"'] & \end{tikzcd}$$

with is the direct sum of the following two representations:

$$\begin{tikzcd} \{0,e_1\} \arrow[r] & \{0,e_1\} & \{0\} \arrow[l] & \{0\} \arrow[r] & \{0,e_2\} & \{0,e_2\} \arrow[l] \\ & \{0,v\} \arrow[u, "v\mapsto e_1"'] & & & \{0\} \arrow[u] & \end{tikzcd}$$

Lemma 9.11. *Let Q be a quiver with underlying graph a tree and V be an indecomposable representation of Q. Then the entries of dimension vector $\underline{\dim}(V)$ are all $0's$ and $1's$. Indecomposable representations of Q correspond to connected subgraphs of Q.*

Proof. To be continued □

9.3 The Hall algebra of $\mathrm{Rep}(Q,\mathbb{F}_1)$

Let $\chi(Q)$ be the set of isomorphism classes of representations in $\mathrm{Rep}(Q,\mathbb{F}_1)$. Set

$$\mathbf{H}_Q = \{f : \chi(Q) \longrightarrow \mathbb{C} \big| \#\mathrm{supp}(f) < \infty\}$$

Define

$$(f \cdot g)(M) = \sum_{L \subset M} f(M/L)g(L), M, L \in \chi(Q)$$

For $M \in \chi(Q)$, write $[M] \in \mathbf{H}_Q$, the delta function supported on the isomorphism class of M. Then

$$[M] \cdot [N] = \sum_{R \in \chi(Q)} \frac{P^R_{M,N}}{a_M \cdot a_N}[R]$$

where

$$P^R_{M,N} = \#\{\text{short exact sequence } 0 \longrightarrow N \longrightarrow R \longrightarrow M \longrightarrow 0\}$$

If we set $\mathbf{P}^R_{M,N} = \frac{P^R_{M,N}}{a_M \cdot a_N}$, then

$$\mathbf{P}^R_{M,N} = \#\{L \subset R \big| L \cong N \text{ and } R/L \cong M\}$$

$\mathbf{H}_Q$ carries a grading by the effective cone $K_0^+(\mathrm{Rep}(Q, \mathbb{F}_1)) \cong \mathbb{Z}^{|I|}_{\geq 0}$ in $K_0(\mathrm{Rep}(Q, \mathbb{F}_1))$, by assigning to $[M]$ the class in $K_0(\mathrm{Rep}(Q, \mathbb{F}_1))$.

Different from the general Hall algebra, for $f \in \mathbf{H}_Q$, we define $\Delta : \mathbf{H}_Q \longrightarrow \mathbf{H}_Q \otimes \mathbf{H}_Q$ by:

$$\Delta(f)([M], [N]) = f([M \oplus N])$$

which makes $\mathbf{H}_Q$ a coalgebra with the counit $\varepsilon([M]) = \delta_{[M],0}$.

Lemma 9.12. *Δ is coassociative, that is, $(\Delta \otimes \mathrm{id}) \circ \Delta = (\mathrm{id} \otimes \Delta) \circ \Delta$.*

Proof. For $[M] \in \chi(Q)$, let $D_n([M]) := \{([A_1], \cdots, [A_n]) \in \chi(Q)^n \big| A_1 \oplus A_2 \oplus \cdots \oplus A_n \cong M\}$. Then

$$\Delta([M]) = \sum_{([A],[B]) \in D_2(M)} [A] \otimes [B]$$

and so

$$(\Delta \otimes \mathrm{id}) \circ \Delta([M]) = \sum_{([A],[B]) \in D_2(M)} \sum_{([A_1],[A_2]) \in D_2(A)} [A_1] \otimes [A_2] \otimes [B]$$

There is an one-to-one corresponding:

$$\{(([A],[B]),([A_1],[A_2])) \big| (A,B) \in D_2(M), (A_1,A_2) \in D_2(A)\} \longleftrightarrow \{([A],[B],[C]) \big| (A,B,C) \in D_3(M)\}$$
$$(([A],[B]),([A_1],[A_2])) \longrightarrow (A_1, A_2, B)$$

Thus,

$$(\Delta \otimes \mathrm{id}) \circ \Delta([M]) = \sum_{([A],[B],[C]) \in D_3(M)} [A] \otimes [B] \otimes [C]$$

Similarly,

$$(\mathrm{id} \otimes \Delta) \circ \Delta([M]) = \sum_{([A],[B],[C]) \in D_3(M)} [A] \otimes [B] \otimes [C]$$

So $(\Delta \otimes \mathrm{id}) \circ \Delta = (\mathrm{id} \otimes \Delta) \circ \Delta$. □

Moreover, we can show that $\mathbf{H}_Q$ is a bialgebra under the multiplication and comultiplication defined above:

Lemma 9.13.

$$\Delta([M] \cdot [N]) = \Delta([M]) \cdot \Delta([N])$$

for $[M], [N] \in \chi(Q)$.

Proof.

$$[M] \cdot [N] = \sum_{[R]} \#\{L \subset R \big| L \cong N \ \&\ R/L \cong M\}[R]$$

whereas

$$\begin{aligned}\Delta([M] \cdot [N]) &= \sum_{[R]} \#\{L \subset R \big| L \cong N \ \&\ R/L \cong M\} \sum_{([A],[B]) \in D_2(R)} [A] \otimes [B] \\ &= \sum_{[A],[B]} \#\{L \subset A \oplus B \big| L \cong N \ \&\ A \oplus B/L \cong M\}[A] \otimes [B]\end{aligned}$$

For RHS,

$$\begin{aligned}\Delta([M])\cdot\Delta([N]) &= \sum_{([A_M],[B_M])\in D_2(M)}\ \sum_{([A_N],[B_N])\in D_2(N)} [A_M]\cdot[A_N]\otimes[B_M]\cdot[B_N] \\ &= \sum_{([A_M],[B_M])\in D_2(M)}\ \sum_{([A_N],[B_N])\in D_2(N)}\ \sum_{[A]}\sum_{[B]} \\ &\quad \#\{L_M^A\subset A\big|L_M^A\cong A_M, A/L_M^A\cong A_M\}\cdot\#\{L_M^B\subset B\big|L_M^B\cong B_M, B/L_M^B\cong B_M\}[A]\otimes[B]\end{aligned}$$

Now, we need to compute the coefficient of $[A]\otimes[B]$. Set $P^A_{A_M,A_N} := \{L_M^A\subset A\big|L_M^A\cong A_M, A/L_M^A\cong A_M\}$ and $P^B_{B_M,B_N} := \{L_M^B\subset B\big|L_M^B\cong B_M, B/L_M^B\cong B_M\}$. Then we can see that

$$\begin{aligned}\{L\subset A\oplus B\big|L\cong M, A\oplus B/L\cong M\} &\overset{1:1}{\longleftrightarrow} \{(L_M^A\subset A, L_M^B\subset B)\big|L_M^A\oplus L_M^B\cong M, A\oplus B/L_M^A\oplus L_M^B\cong N\} \\ &= \bigsqcup_{(A_M,B_M)\in D_2(M)}\ \bigsqcup_{(A_N,B_N)\in D_2(N)} P^A_{A_M,A_N}\times P^B_{B_M,B_N}\end{aligned}$$

where the first map is given by $(L\subset A\oplus B)\longmapsto(L\cap A\subset A, L\cap B\subset B)$. Thus, the coefficient of $\Delta([M])\cdot\Delta([N])$ is just

$$\#\bigsqcup_{(A_M,B_M)\in D_2(M)}\ \bigsqcup_{(A_N,B_N)\in D_2(N)} P^A_{A_M,A_N}\times P^B_{B_M,B_N} = \#\{L\subset A\oplus B\big|L\cong M, A\oplus B/L\cong M\}$$

which proves our result. □

Now, we show that $\mathbf{H}_Q$ is a graded bialgebra with $H_0\cong k$. So it is connected. By the following theorem, $\mathbf{H}_Q$ is a Hopf algebra.

Theorem 9.14 (Milnor-Moore reference?)**.** *Every connected graded bialgebra over a field automatically admits a unique antipode, hence is a Hopf algebra.*

$\mathbf{H_Q}$ is automatically, a Hopf algebra. Moreover, Milnor and Moore prove that

Theorem 9.15 (Milnor–Moore,reference?)**.** *Let k be a field of character 0 and H be a graded and connected, cocommutative Hopf algebra over k. Set the space of primitive elements*

$$\mathrm{P}(H) := \{x\in H \mid \Delta(x) = x\otimes 1 + 1\otimes x\}$$

which is a Lie algebra under the commutator bracket $[x,y] := xy - yx$ and the canonical algebra homomorphism

$$U(\mathrm{P}(H)) \longrightarrow H$$

is an isomorphism of Hopf algebras.

For $\mathbf{H}_Q$ is a graded, connected cocommutative Hopf algebra, it is isomorphic to $\mathbf{U}(\mathfrak{n}_Q)$, for $\mathrm{n}_Q = P(\mathbf{H}_Q)$. From the definition of the coproduct, $[M]\in\mathbf{H}_Q$, $[M]\in\mathbf{H}_Q$ is primitive if and only if M is indecomposable. Thus,

$$\mathfrak{n}_Q = \{[M]\big|M \text{ indecomposable}\}$$

with $[M,N] = [M]\cdot[N] - [N]\cdot[M]$.

We cam also extend $\mathbf{H}_Q$ by adjointing to it a degree zero piece coming from $K_0(\mathrm{Rep}(Q,\mathbb{F}_))$. We define

$$\mathfrak{h}_Q = \mathrm{Hom}_{\mathbb{Z}}(K_0(\mathrm{Rep})(Q,\mathbb{F}_1),\mathbb{C})$$

and so $\mathfrak{h}_Q^* \cong K_0(\mathrm{Rep}(Q, \mathbb{F}_1)) \otimes_{\mathbb{Z}} \mathbb{C}$ and $\mathrm{Sym}(\mathfrak{h}_Q) \cong \mathbb{C}[[S_i]^{\pm} \big| i \in I]$. Let now, $\widetilde{\mathbf{H}}_Q := \mathrm{Sym}(\mathfrak{h}_Q) \otimes_{\mathbb{C}} \mathbf{H}_Q$, equipped with

$$[Z, f] := Z(\alpha) f \text{ for } Z \in \mathfrak{h}_Q \text{ and } f \in \mathbf{H}_Q[\alpha]$$

With twisted multiplication, $\widetilde{\mathbf{H}}_Q$ is isomorphic to $U(\mathfrak{b}_Q)$ with $\mathfrak{b}_Q := \mathfrak{h}_Q \oplus \mathfrak{n}_Q$, where the Lie bracket between $\mathfrak{h}_Q$ and $\mathfrak{n}_Q$ given above. Moreover, we extend the coproduct of $\mathbf{H}_Q$ to $\widetilde{\mathbf{H}}_Q$ by

$$\Delta(Z) = Z \otimes 1 + 1 \otimes Z \in \widetilde{\mathbf{H}}_Q \otimes \widetilde{\mathbf{H}}_Q$$

Recall that for a given quiver Q, we can construct a Kac-Moody algebra $\mathfrak{g}$ associated to Q 5.13. What's the relation between the Borel subalgebra $\mathfrak{b}$ of $\mathfrak{g}$ and $\mathfrak{b}_Q$?

Theorem 9.16. *The assignment $e_i \longmapsto [S_i]$, $h_i \longmapsto Z_i$ extneds to a Hopf algebra homomorphism*

$$\rho : U(\mathfrak{b}) \longrightarrow \widetilde{\mathbf{H}}_Q$$

where Z_i satisfies $Z_i(\alpha_j) = a_{ij}$.

Proof. For $U(\mathfrak{b})$, it is a Lie algebra generated by $e_i, h_i, i \in I$ with relations

$$[h_i, e_j] = a_{ij} e_j, [h_i, h_j] = 0, \mathrm{ad}^{1-a_{ij}}(e_i) e_j = 0$$

where $\mathrm{ad}^{1-a_{ij}}(e_i) e_j = 0$ corresponds to

$$\sum_{s=0}^{1-a_{ij}} (-1)^s \binom{1-a_{ij}}{s} e_i^s e_j e_i^{1-a_{ij}-s} = 0$$

with coproduct $\Delta : x \mapsto x \otimes 1 + 1 \otimes x$ for $x = e_i, h_i, i \in I$. Hence, to check that ρ is a morphism of Hopf algebras, we need to check

$$[Z_i, Z_j] = 0, [Z_i, S_j] = a_{ij} S_j, \mathrm{ad}^{1-a_{ij}}(S_j) S_i = 0$$

and

$$\Delta([S_i]) = [S_i] \otimes 1 + 1 \otimes [S_i], \Delta(Z_i) = Z_i \otimes 1 + 1 \otimes Z_i$$

Above all relations, we only need to verify the Serre relations, $\mathrm{ad}^{1-a_{ij}}(S_j) S_i = 0$, which is the same as the proof of 6.2. □

Remark. Restricting ρ to $U(\mathfrak{n}_+)$, we obtain a Hopf algebra homomoprhism:

$$\rho' : U(\mathfrak{n}_+) \longrightarrow \mathbf{H}_Q$$

Since both algebras are enveloping algebras, by the universal properties of envelopong algebra, ρ, ρ' are determined by the corresponding Lie algebra homoprhism $\rho_{Lie} : \mathfrak{b}_+ \longrightarrow \mathfrak{b}_Q$.

When ρ is an isomorphism at this case? Different from the general case, only when the underlying graph of Q is a Dynkin diagram of type A, ρ is an isomorphism.

Example 9.17. Consider D_4, as we have seen, the root system is

$$\{\alpha_1, \alpha_2, \alpha_3, \alpha_4\} \cup \{\alpha_1 + \alpha_2, \alpha_2 + \alpha_3, \alpha_2 + \alpha_4\} \cup \{\alpha_1 + \alpha_2 + \alpha_4, \alpha_1 + \alpha_2 + \alpha_3, \\ \alpha_2 + \alpha_3 + \alpha_4\} \cup \{\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4, \alpha_1 + 2\alpha_2 + \alpha_3 + \alpha_4\}$$

while the indecomposables are

$$\{S_1, S_2, S_3, S_4\} \cup \{R_{12}, R_{23}, R_{24}\} \cup \{R_{124}, R_{123}, R_{234}\} \cup \{R_{1234}\}$$

We can see that the indecomposable corresponding to the maximal root is missing over $\mathbb{F}_1$: We denote by e_i, e_{ij}, e_{ijk} basis vectors for one-dimensional root spaces $\alpha_i, \alpha_i + \alpha_j, \alpha_i + \alpha_j + \alpha_k$. We have the relations: $[e_i, e_j] = e_{ij}$ and similarly for e_{ijk}. Then

$$\rho(e_i) = S_i, \rho(e_{ij}) = R_{ij}, \rho(e_{ijk}) = R_{ijk}, \rho(e_{1234}) = R_{1234}$$

and $\rho(e_{12^234}) = \rho([e_{1234}, e_2]) = [R_{1234}, S_2] = 0$. Thus, ρ is not an isomorphism.

Theorem 9.18. *When Q is a quiver of type A, then the homomorphism $\rho : \mathbf{U}(\mathfrak{b}) \longrightarrow \widetilde{\mathbf{H}}_Q$ is an isomorphism.*

Proof. It is clear that $\mathfrak{h}$ maps isomorphically to $\mathfrak{h}_Q$, so it suffices to check that

$$\rho' : \mathfrak{n}_+ \longrightarrow \mathfrak{n}_Q$$

is an isomorphism. Since $\mathfrak{n}_+$ and $\mathfrak{n}_Q$ have the same dimension, it suffices to show that ρ' is surjective, that is, all $[M]$ with M indecomposable lie in the image of ρ'. We will prove this by induction.

By the definition of ρ, the simples S_j are in the image of ρ', and suppose that V is an indecomposable in $\mathrm{Rep}(Q, \mathbb{F}_1)$ such that if $W \in \mathrm{Rep}(Q, \mathbb{F}_1)$ is indecomposable and $\dim W < \dim V$, then $W \in \mathrm{Im}(\rho')$. The support of V is connected, and so it must be a constructive chain of vertices:

$$V: \overset{1}{\bullet} \dashrightarrow \overset{i}{\bullet} \longrightarrow \overset{i+1}{\bullet} \dashrightarrow \overset{j-1}{\bullet} \longrightarrow \overset{j}{\bullet} \dashrightarrow \overset{n}{\bullet}$$

Let V' be

$$\overset{1}{\bullet} \dashrightarrow \overset{i}{\bullet} \dashrightarrow \overset{i+1}{\bullet} \longrightarrow \overset{i+1}{\bullet} \dashrightarrow \overset{j-1}{\bullet} \longrightarrow \overset{j}{\bullet} \dashrightarrow \overset{n}{\bullet}$$

Then by 9.11, V' is also an indecomposable, and therefore $[V'] \in \mathrm{Im}(\rho')$ by induction hypothesis. Since

$$[S_i] \cdot [V'] = [V] + [S_i \oplus V], \ [V'] \cdot [S_i] = [S_i \oplus V']$$

$[V] = [[S_i], [V']]$ lies in the image of $\mathrm{im}(\rho')$. Thus, ρ' is a surjection by induction. $\square$

10 Jordan quiver

Now, let's consider quivers with loops, where the simplest example is the Jordan quiver. The Jordan quiver is the oriented graph Q_0 with a single vertex i and a single loop $h : i \longrightarrow i$:

$$\begin{tikzcd} \bullet_{i} \arrow[loop above, "h"] \end{tikzcd}$$

We will study the Hall algebra $\mathbf{H}_{Q_0}$ of Q_0 and find the relation between $\mathbf{H}_{Q_0}$ and the ring of symmetry functions.

10.1 Hall algebra of Jordan quiver

Let k be a field. By 5.3, a representation of Q_0 over k is a pair (V, x) consisting of a finite-dimension k-vector space V and a k-linear map $x : V \longrightarrow V$.

A morphism between two representations (V, x) and (V', x') of Q_0 is a linear map $f : V \longrightarrow V'$ such that

$$\begin{tikzcd} V \arrow[r, "x"] \arrow[d, "f"] & V \arrow[d, "f"] \\ V' \arrow[r, "x'"] & V' \end{tikzcd}$$

Thus, the category $\mathrm{Rep}_k(Q_0)$ is equivalent to the category of finite-dimensional modules over $k[x]$.

We will be interested in the full subcategory of those representation (V, x) for which x is a nilpotent endomorphism of V. We denote this categroy by $\mathrm{Rep}_k^{nil}(Q_0)$.

For nilpotnet endomorphisms in linear spaces we have the following results:

Theorem 10.1 (Jordan)**.** *The following hold:*

i). The object $S = (k, 0)$ is the only simple object of $\mathrm{Rep}_k^{nil}(Q_0)$;

ii). We have $\mathrm{Hom}(S, S) = k$ and $\mathrm{Ext}^1(S, S) = k$;

iii). For any $n \in \mathbb{N}$, there exists a unique indecomposable object I_n of length n in $\mathrm{Rep}_k^{nil}(Q_0)$, given by

$$x = \begin{pmatrix} 0 & 1 & 0 & \cdots & 0 \\ 0 & 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & 1 \\ 0 & 0 & 0 & \cdots & 0 \end{pmatrix}$$

Remark. i). Since $S = (k, 0)$ is the only simple object of $\mathrm{Rep}_k^{nil}(Q_0)$, $K_0(\mathrm{Rep}_k^{nil}(Q_0)) = \mathbb{Z}$;

ii). Since $\mathrm{Rep}_k(Q_0)$ is Krull-Schmit, any (V, x) in $\mathrm{Rep}_k^{nil}(Q_0)$ is isomorphic to a direct sum $I_1^{\oplus l_1} \oplus I_2^{\oplus l_2} \oplus \cdots \oplus I_r^{\oplus l_r}$ for some r and $l_1, \cdots, l_r$. Thus, $\chi_{\mathrm{Rep}_k^{nil}(Q_0)}$ is isomorphic to the set Π of all partitions, via:

$$\Pi \longrightarrow \chi_{\mathrm{Rep}_k^{nil}(Q_0)}$$
$$\lambda = (\lambda_1, \cdots, \lambda_r) \longmapsto I_\lambda := I_{\lambda_1} \oplus I_{\lambda_2} \oplus \cdots \oplus I_{\lambda_r}$$

Example 10.2. We're going to compute some Hall numbers in H_{Q_0}, i.e.

$$P^{\nu}_{\mu,\lambda} := \#\{L \subset I_v \big| L \cong I_\lambda \text{ and } I_v/L \cong I_\mu\}$$

i). $P^{(1^2)}_{(1),(1)}$ is just the number of subrepresentations of $S^{\oplus 2} = (k^2, 0)$, which is just the number of 1-dimensional subspaces of k^2. Thus, $P^{(1^2)}_{(1),(1)} = |\mathbb{P}(k)| = q + 1$.

ii). $P^{(2)}_{(1),(1)} = \#\{L \subset I_2 \big| L \cong S \text{ and } I_2/L \cong S\}$. Note that

$$I_2 = (k^2, x = N(2) = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix})$$

which contains only 1-subrepresentation isomorphic to S. Thus, $P^{(2)}_{(1),(1)} = 1$.

iii). $P^{(2,1)}_{(2),(1)} = \#\{L \subset I_2 \oplus S \big| S = L \text{ and } I_2 \oplus S/L \cong I_2\}$. Note

$$I_2 \oplus S = (k^3, x = \begin{pmatrix} 0 & 1 & \\ 0 & 0 & \\ & & 0 \end{pmatrix})$$

Write $k^3 = \langle e_1, e_2, e_3 \rangle$ with $x(e_2) = e_1, x(e_1) = 0, x(e_3) = 0$. Then $\ker x = \langle e_1, e_3 \rangle$ and so L must be a subrepresentation of $\ker x$ of dimension 1. For $\operatorname{im} x = \langle e_1 \rangle \subset \ker x$. If $I_2 \oplus S/L \neq S^{\oplus 2}$, we need $\operatorname{im} x \not\subset L$. Thus, L can only be of the form $\langle ae_1 + e_3 \rangle$ with $a \in k$. Moreover, when L is of such forms, $I_2 \oplus S/L \cong I_2$. Thus, $P^{(2,1)}_{(2),(1)} = |k| = q$.

iv). $P^{(2,1)}_{(1),(2)} = \#\{L \subset I_2 \oplus S \big| L \cong I_2, I_2 \oplus S/L \cong S\}$. We set everything the same as iii). Then since $x|_L$ is non-trivial, $\ker x \not\subseteq L$ and $\operatorname{im} x \subset L$ since we need $I_2 \oplus S/L \cong S$. Thus, L is of the form

$$\langle e_1, e_2 + be_3 \rangle$$

with $b \in k$. Then $x|_L \cong N(2)$ and so $P^{(1,2)}_{(1),(2)} = |k| = q$.

v). $P^{(1^t)}_{(1^{t-r}),(1^r)} = \#\{L \subset S^{\oplus t} \big| L \cong S^{\oplus r}, S^{\oplus t}/L \cong S^{\oplus (t-r)}\}$, as we discuss at the first example, it counts the number of r-dimensional subspaces of $k^{\oplus t}$, which is just $\#|\mathrm{Gr}(r,t)|$. So

$$P^{(1^t)}_{(1^{t-r}),(1^r)} = \binom{t}{r}_q$$

vi). $P^{(t)}_{(t-r),(r)} = \#\{L \subset I_t \big| L \cong I_r, I_t/L \cong I_{t-r}\}$. By general linear algebra, there exists only one subspace of $k^{\oplus t}$, L such that

$$(L, x|_L) \cong (k^{\oplus (t-r)}, N(t-r))$$

Thus, $P^{(t)}_{(t-r),(r)} = 1$.

Now, consider $\mathrm{Rep}^{nil}(Q_0, \mathbb{F}_0)$. The simple object in $\mathrm{Rep}^{nil}(Q_0, \mathbb{F}_0)$ is

$$S = (\{0, v_1\}, x = 0)$$

At this case, just by 9.4, any $(V, x) \in \mathrm{Rep}^{nil}(Q_0, \mathbb{F}_1)$ can be written as a direct sum

$$N_1^{\oplus n_1} \oplus N_2^{\oplus n_2} \oplus \cdots \oplus N_r^{\oplus n_r}$$

where $N_n = (\mathbf{m}, N_m)$. So, we can also identify $\chi_{\mathrm{Rep}^{nil}(Q_0, \mathbb{F}_1)}$ with Π, the set of partitions.

Example 10.3. We also compute some Hall numbers over $\mathbf{H}_{Q_0}$:

i). $P^{(1^2)}_{(1),(1)} = 2$ since by direct counting, there are only two subspace of $(\{0, v_1, v_2\}, x = 0)$, say $\{0, v_1\}, \{0, v_2\}$ such that $(L, x|_L) \cong S$ and $S^{\oplus 2}/L \cong S$.

ii). $P^{(2)}_{(1),(1)} = \#\{L \subset N_2 \big| L \cong S, N_2/L \cong S\}$. Note that

$$N_2 = (\{0, 1, 2\}, (x : 2 \to 1, 1 \to 0))$$

Then $\ker x = (\{0, 1\}, (x : 1 \longrightarrow 0)) \cong S$. Thus, $P^{(2)}_{(1),(1)} = 1$.

iii). $P^{(2,1)}_{(2),(1)} = \#\{L \cong N_2 \oplus S \big| L \cong S, N \oplus S/L \cong N_2\}$, where

$$N \oplus S = (\{0,1,2,3\}, (x : 3 \to 0, 2 \to 1, 1 \to 0)$$

$\ker x = \{0,1,3\}$ and $\operatorname{im} x = \{0,1\}$. So the only element satisfying the condition above is $(\{0,3\}, (x : 3 \to 0))$.

iv). $P^{(1^t)}_{(1^{t-r}),(1^r)} = \#\{L \subset S^{\oplus t} \big| L \cong S^{\oplus r}, S^{\oplus t}/L \cong S^{\oplus(t-r)}\}$, which counts the number of t-element set in r-element set. Thus, $P^{(1^t)}_{(1^{t-r}),(1^r)} = \binom{t}{r}$.

v). Just as before ,$P^{(t)}_{(t-r),(r)} = 1$.

By 9.3, there exists an isomorphism

$$\mathbf{H}_{Q_0} \cong \mathbf{U}(\mathfrak{n}_{Q_0})$$

where $\mathfrak{n}_{Q_0}$ is the Lie algebra spanned by $[N_i]$ with $i \geq 1$. By the computation of Hall number,

$$[N_i] \cdot [N_j] = [N_i \oplus N_j] + [N_{i+j}]$$

This implies $\mathfrak{n}_{Q_0}$ is abelian and so is $\mathbf{H}_{Q_0}$.

Now, we're going to use this condition to identify $\mathbf{H}_{Q_0}$ with Λ, the ring of symmetry functions. But before that, we first introduce some basic notations related to symmetry functions.

10.2 Symmetry functions

Let $x_1, \cdots, x_n$ be independent indeterminates. There is an action of S_n over $\mathbb{Z}[x_1, \cdots, x_n]$:

$$\begin{aligned} S_n \times \mathbb{Z}[x_1, ..., x_n] &\longrightarrow \mathbb{Z}[x_1, ..., x_n] \\ (\sigma, f(x_1, ..., x_n)) &\longmapsto f(x_{\sigma(1)}, ..., f_{\sigma(n)}) \end{aligned}$$

Then we just set $\Lambda_n = \mathbb{Z}[x_1, ..., x_n]^S_n$. If we adjoint another indeterminate x_{n+1}, we can form $\Lambda_{n+1} = \mathbb{Z}[x_1, ..., x_n, x_{n+1}]^{S_{n+1}}$, and we have a surjection

$$\begin{aligned} p_{n+1} : \Lambda_{n+1} &\longrightarrow \Lambda_n \\ f(x_1, \cdots, x_n, x_{n+1}) &\longrightarrow f(x_1, \cdots, x_n, 0) \end{aligned}$$

Moreover, if $f \in \Lambda_n$, we may write $f = \sum_{r \geq 0} f^{(r)}$, where $f^{(r)}$ is the homogeneous component of f of degree r. Each $f^{(r)}$ is itself symmetry, and so Λ_n is a grading ring:

$$\Lambda_n = \oplus_{r \geq 0} \Lambda^r_n$$

where Λ^r_n is the additive group of homogeneous symmetry polynomials of degree r in $x_1, ..., x_n$. Under the mapping p_{n+1}, there exists a map

$$\Lambda^r_{n+1} \longrightarrow \Lambda^r_n$$

which is surjective for all $r \geq 0$ and bijective if and only if $r \leq n$. Hence, we can pass to the limit

$$\Lambda^r = \varprojlim \Lambda^r_n$$

for each $r \geq 0$ and let

$$\Lambda = \oplus_{r \geq 0} \Lambda^r$$

By the definition of inverse limit, an element of Λ^r is a sequence $(f_n)_{n \geq 0}$ such that $f_n(x_1, ..., x_n) = f_{n+1}(x_1, ..., x_n, 0)$. Thus, we regard f_n as the partial sum of an infinite series of monomial of degree r in infinite many indeterminates $x_1, x_2, ...$ Now, we will introduce some important elements in Λ, *monomial symmetry functions, elementary symmetry functions* and *complete symmetry functions.*

Definition 10.4. Let $\lambda = (\lambda_1, \lambda_2, ...)$ be a partition. The monomial function m_λ is defined to be

$$m_\lambda = \sum_{\sigma \in S} x_{\sigma(1)}^{\lambda_1} x_{\sigma(2)}^{\lambda_2} \cdots$$

Example 10.5. For $\lambda_1 = (2,1), \lambda_2 = (2,2,1)$,

$$m_{\lambda_1} = \sum_{i \neq j} x_i^2 x_j; \; m_{\lambda_2} = \sum_{i \neq j, i \neq k, j \neq k} x_i^2 x_j^2 x_k$$

As λ runs through the partitions of length $\leq n$,

$$m_\lambda(x_1, ..., x_n) = \sum_{\sigma \in S_n} x_{\sigma(1)}^{\lambda_1} x_{\sigma(2)}^{\lambda_2} \cdots x_{\sigma(n)}^{\lambda_n}$$

form a $\mathbb{Z}$-basis of Λ_n. So as λ runs through all partitions, m_λ form a $\mathbb{Z}$-basis of Λ.

Definition 10.6. The r-th elementary symmetry function e_r is defined to be

$$e_r = m_{(1^r)} = \sum_{i_1 < i_2 < \cdots < i_r} x_{i_1} x_{i_2} \cdots x_{i_r}$$

For $r = 0$, we define e_0 to be 1.

It is easy to see that e_r admit a generating function:

$$E(t) = \sum_{r \geq 0} e_r t^r = \Pi_{i \geq 1}(1 + x_i t)$$

Lemma 10.7. $\Lambda = \mathbb{Z}[e_1, e_2, \cdots]$ *and* $e_1, e_2, \cdots$ *are algebraically independent over* $\mathbb{Z}$.

Proof. As $\{m_\mu\}$ is a basis of Λ, we show that $\{e_\lambda\}$ is also a basis via representing them by m_μ.
Write $\square$

Appendix A:Representations of Quantum Groups

10.3 Irreducible Representations of $\mathbf{U}_q(\mathfrak{sl}_2)$

Any finite dimensional irreducible $\mathbf{U}_q(\mathfrak{sl}_2)$-module V is generated by a highest weight vector v of weight ϵq^d where $\epsilon = \pm 1$ and $d = \dim V - 1$.

Let $\{v = v_0, v_1, ..., v_d\}$ be a basis of V. We denote this representation by V_d. Then the action of $\mathbf{U}_q(\mathfrak{sl}_2)$ on V_d is given by

$$\begin{aligned} K^{\pm} v_m &= q^{\pm(d-2m)} v_m \\ E v_m &= [d-i+1]_q \, v_{m-1} \\ F v_m &= [m+i]_q \, v_{m+1} \end{aligned}$$

See [[Jan96] Chapter 2]

References

[Har77] Robin Hartshorne. *Algebraic Geometry*. Vol. 52. Graduate Texts in Mathematics. New York-Heidelberg: Springer-Verlag, 1977. ISBN: 978-0-387-90244-9. DOI: 10.1007/978-1-4757-3849-0.

[Jan96] Jens Carsten Jantzen. *Lectures on Quantum Groups*. Vol. 6. Graduate Studies in Mathematics. Providence, RI: American Mathematical Society, 1996. ISBN: 978-0-8218-0478-0.